A Consumer Search Explanation for Hidden Fees

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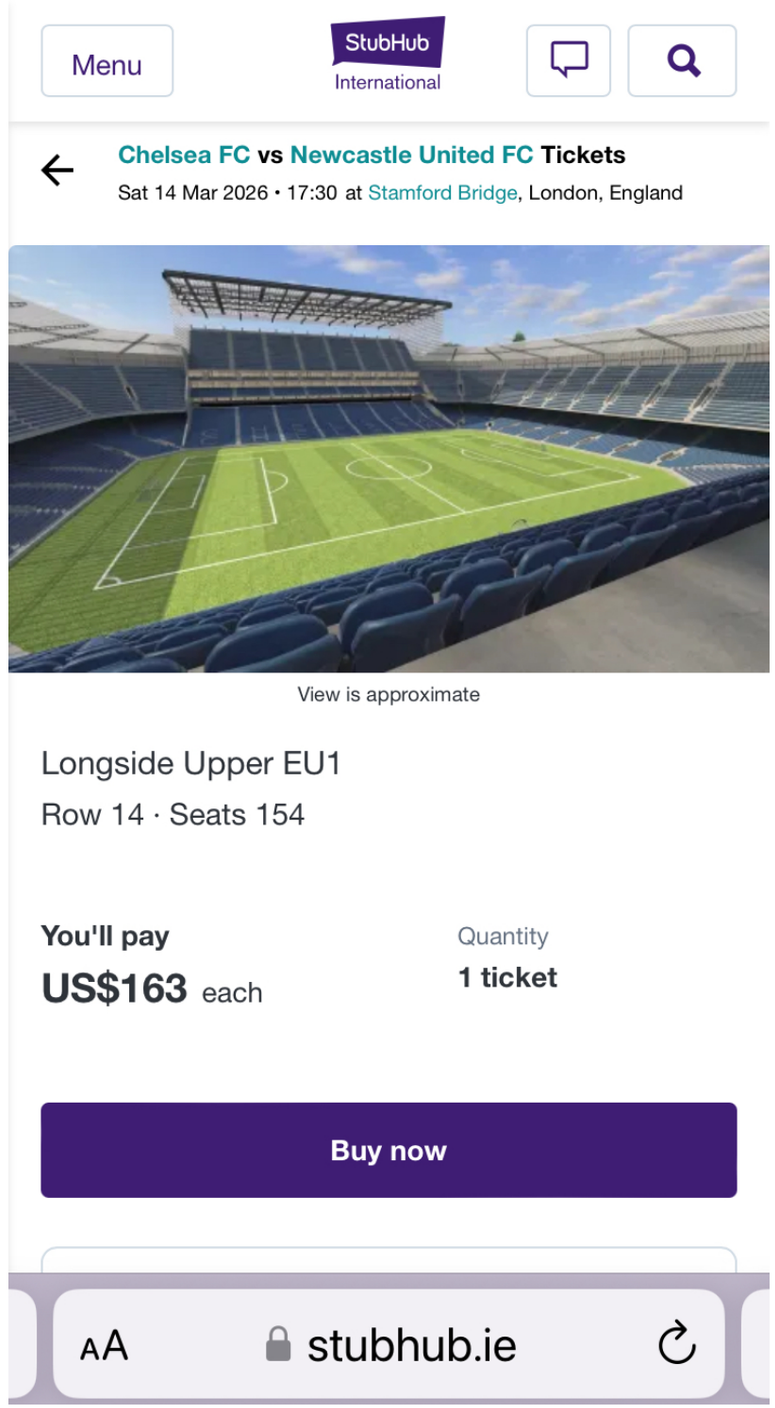
An example of hidden fee

StubHub:



An example of hidden fee

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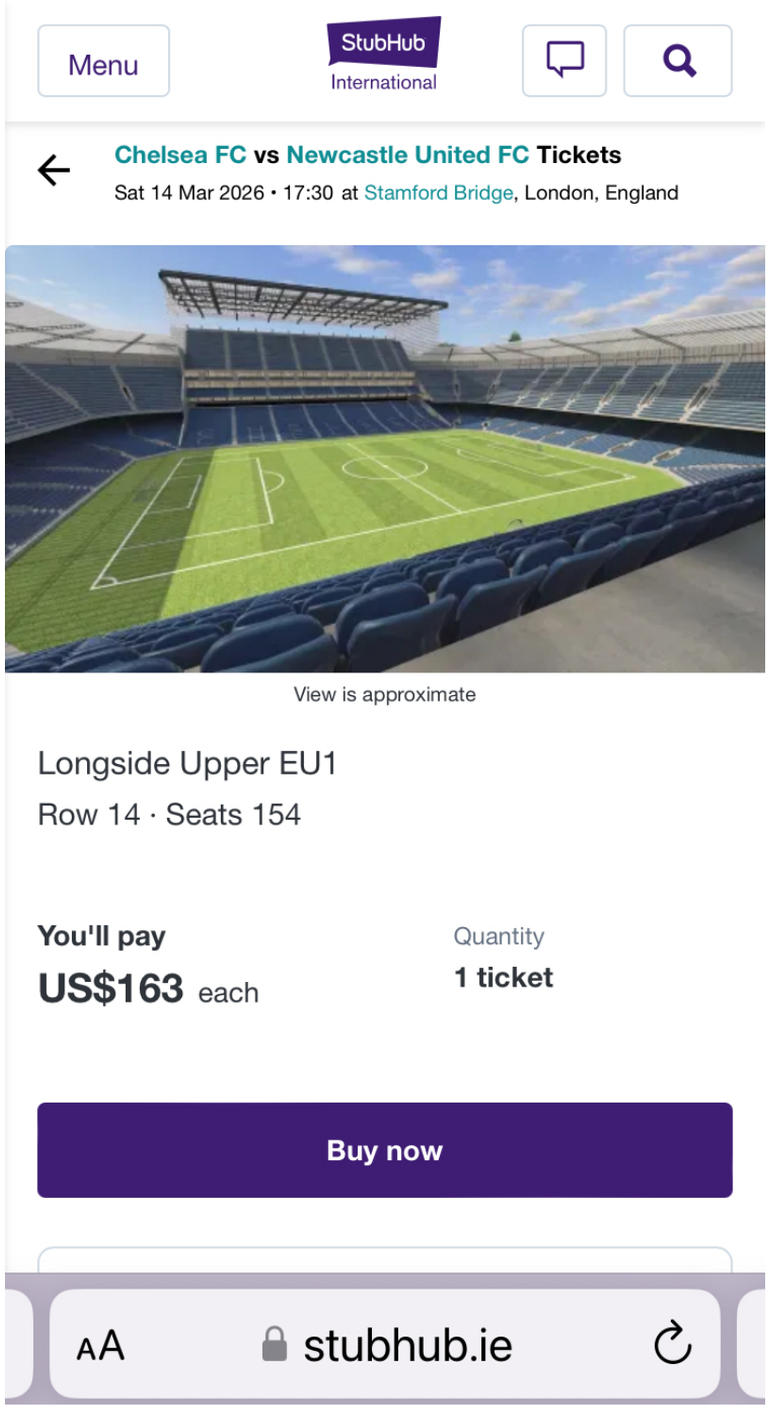
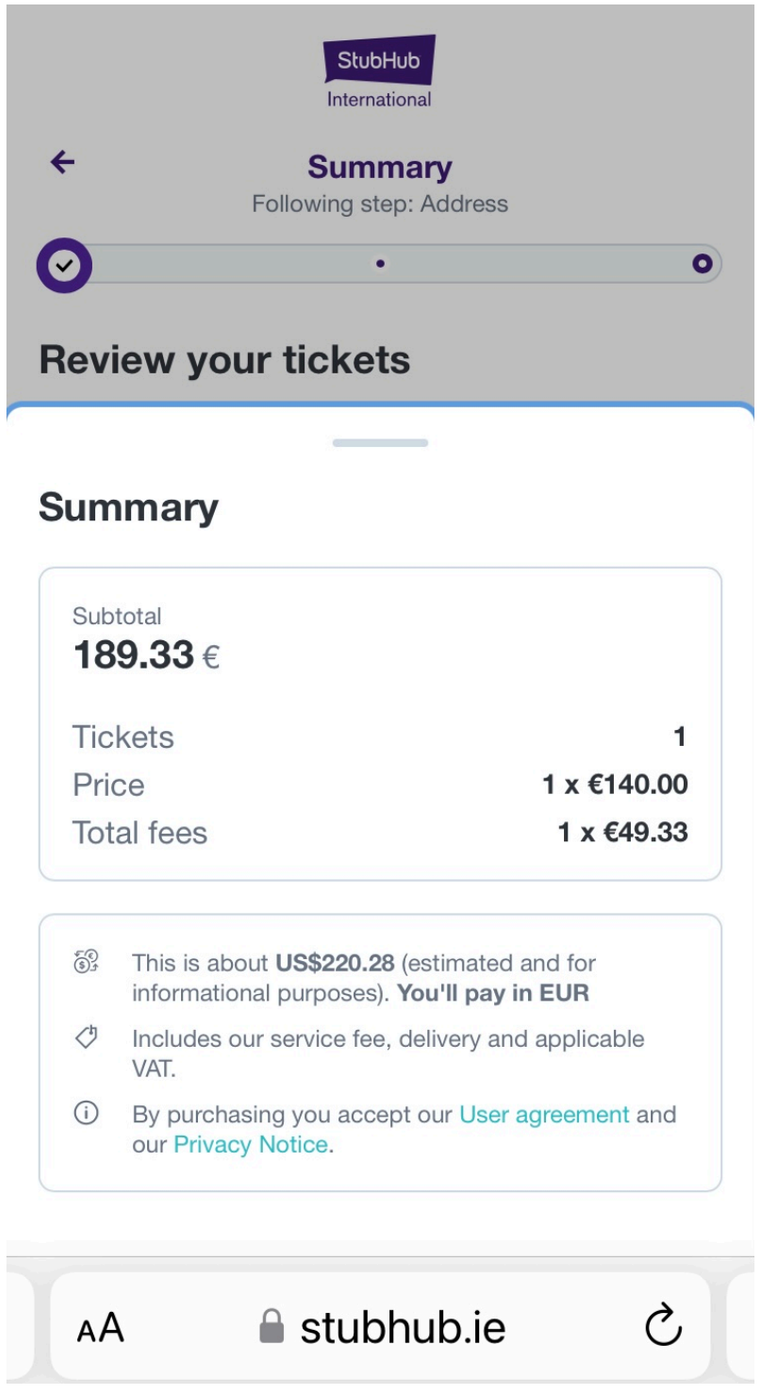




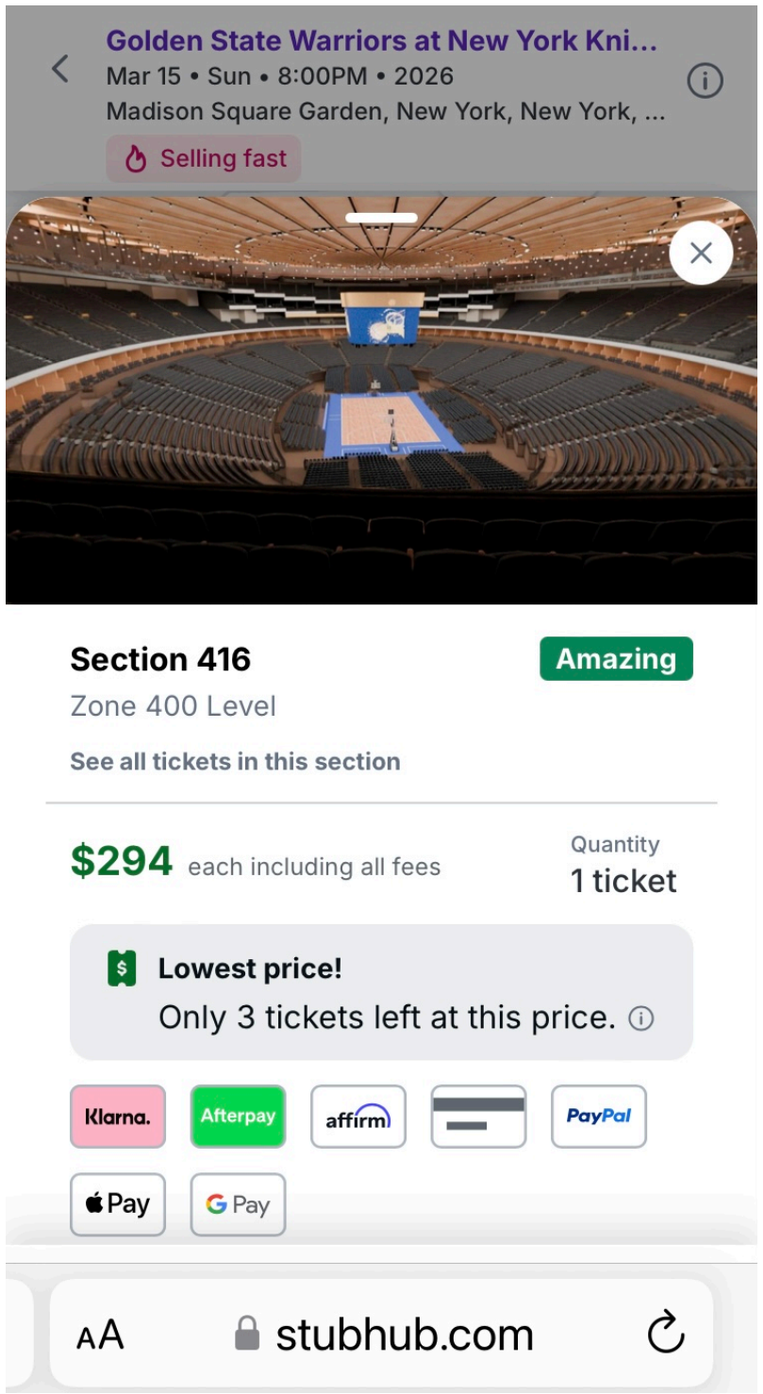
Hidden fee is impactful

- Blake et. al (2021): hidden fee leads to
  - **21%** more money spent
  - **14%** higher purchase likelihood
- Many other studies find that hidden fees affect demand
  - Chetty, Looney, and Kroft, 2009; Brown, Hossain, and Morgan, 2010; Feldman and Ruffle, 2015; Feldman, Goldin, and Homonoff, 2018; Bradley and Feldman, 2020

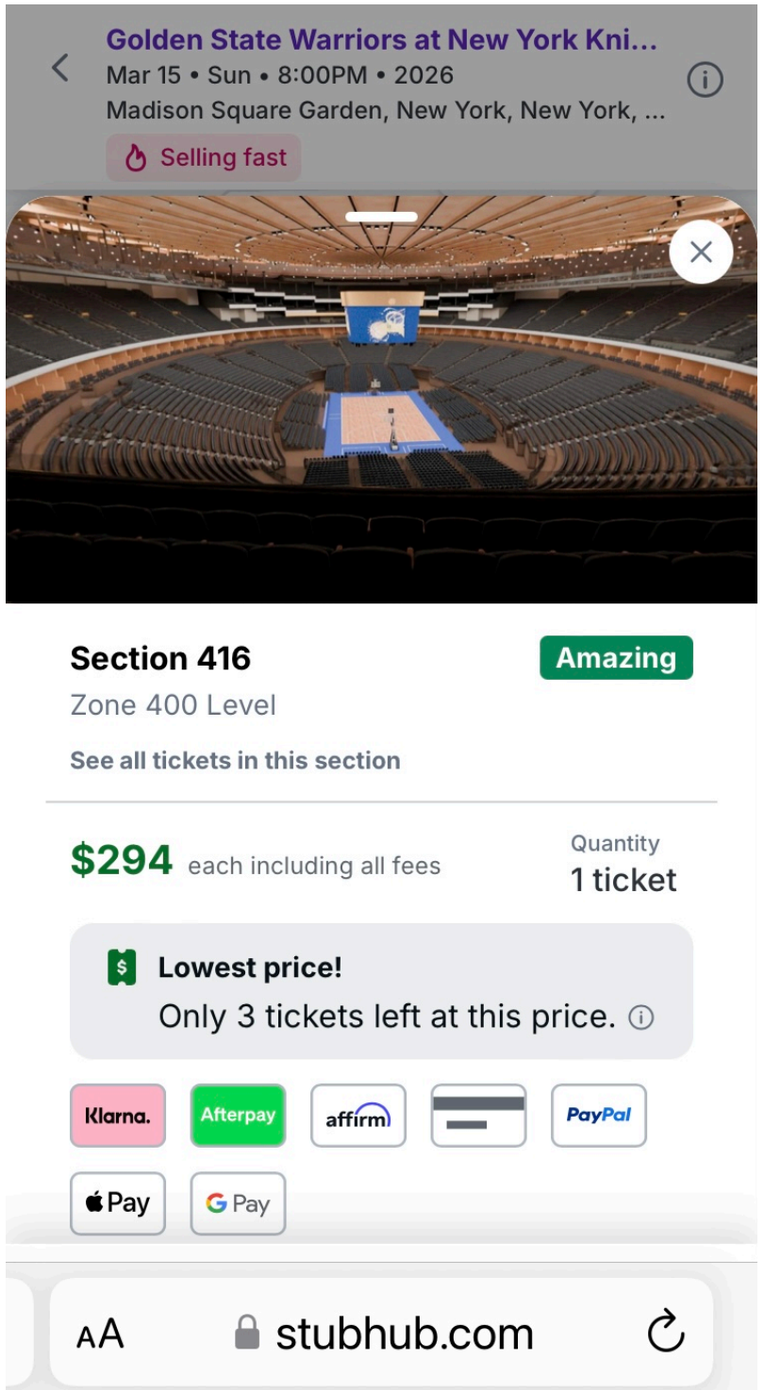
Regulations

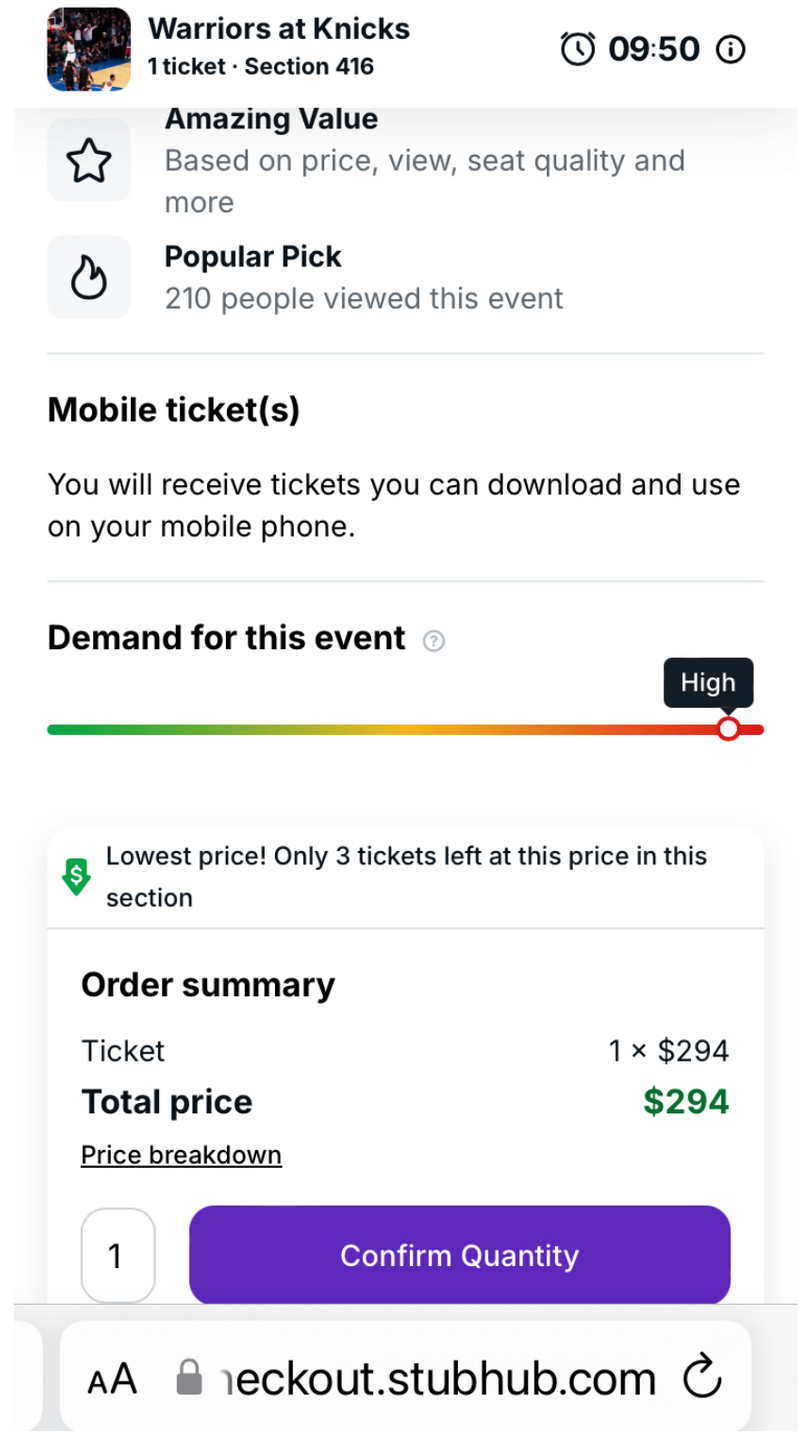
- **New York (2022):** "all-in pricing" law for live event tickets
- **California (2024):** "Hidden Fees Statute" (SB 478) for most consumer transactions
- **Federal Trade Commission (2025):** "Junk Fees Rule” for live-event tickets and short-term lodging
- **European Union (Payment Services Regulation):** In negotiation for international payments

Regulations



Regulations





This paper

- Provides an explanation for why hidden fees benefit the firm by considering **consumer search**

- Examines the policy implications of the (dis)use of hidden fees?

Related Research

- Shrouding can persist in equilibrium under consumer unawareness: Ellison 2005, Gabaix and Laibson 2006

- Empirical support: Jin, Luca, and Martin, 2021; Montero and Sheth, 2021; Brown, Hossain, and Morgan, 2010; Brown, Camerer, and Lovallo, 2012; Sheth, 2021; Sah and Read, 2020

- How consumer unawareness shapes equilibrium pricing in markets with add-ons: Gabaix and Laibson 2006; Armstrong and Vickers, 2012; Johnen and Somogyi, 2024; Heidhues, Koszegi, and Murooka, 2016; Kosfeld and Schuwer, 2017; Shulman and Geng, 2013; Geng, Tan, and Wei, 2018; Erat and Bhaskaran, 2012

Related Research

- **Optional add-ons**: remain unnoticed when the base product is purchased
  - Standard shrouded-attributes mechanism: consumers purchasing the base product without fully accounting for add-on prices
- **Mandatory fees:** disclosed before the transaction is finalized
  - Hard to reconcile with the standard shrouded-attributes logic

Basic consumer search model (Branco et al. 2012)

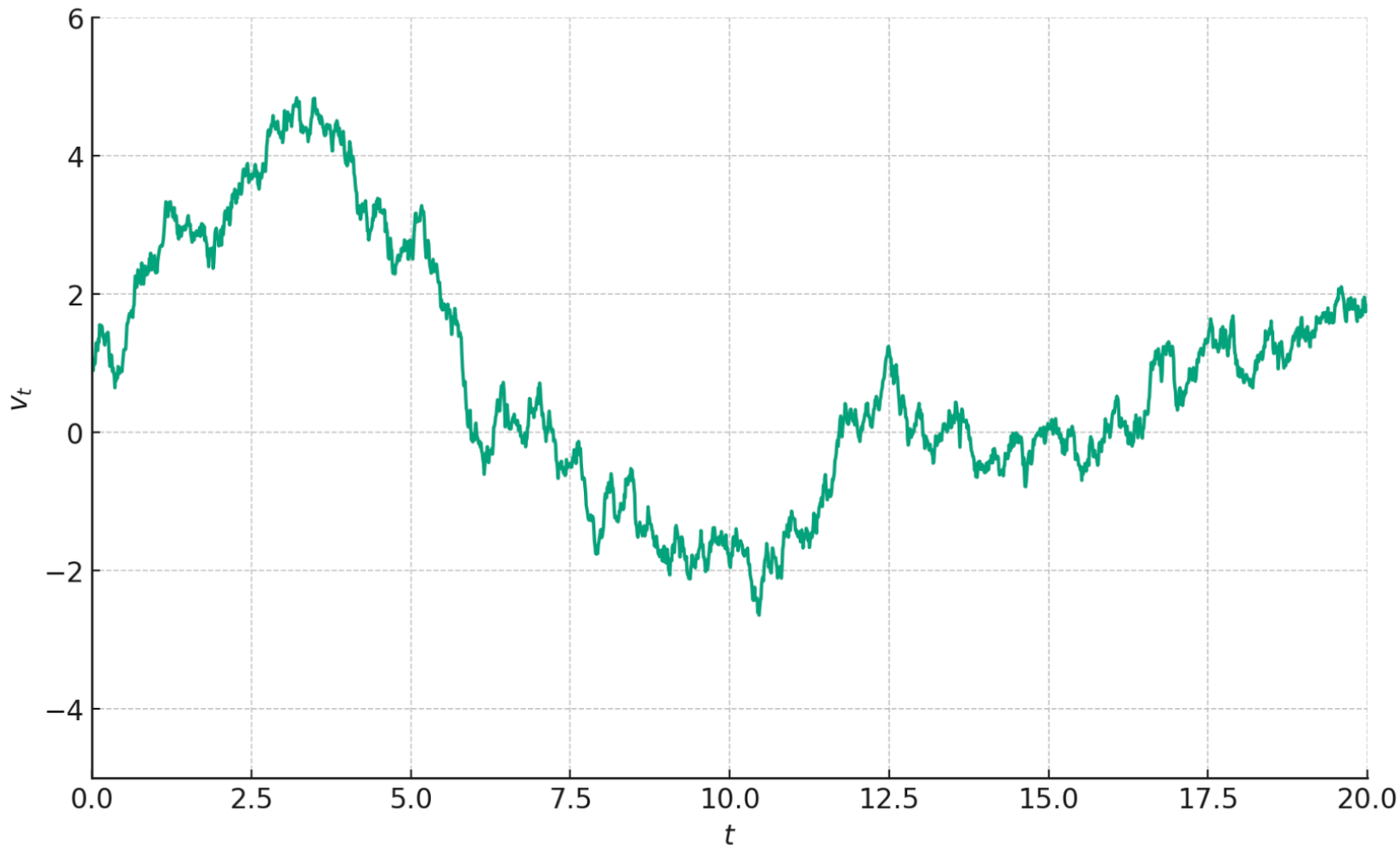
- A firm offers a product with a marginal cost of m and price p
- A consumer decides whether to buy it
- The consumer can search for information before making a decision

  initial valuation: v_0 (common knowledge)

  search cost: c per period of time

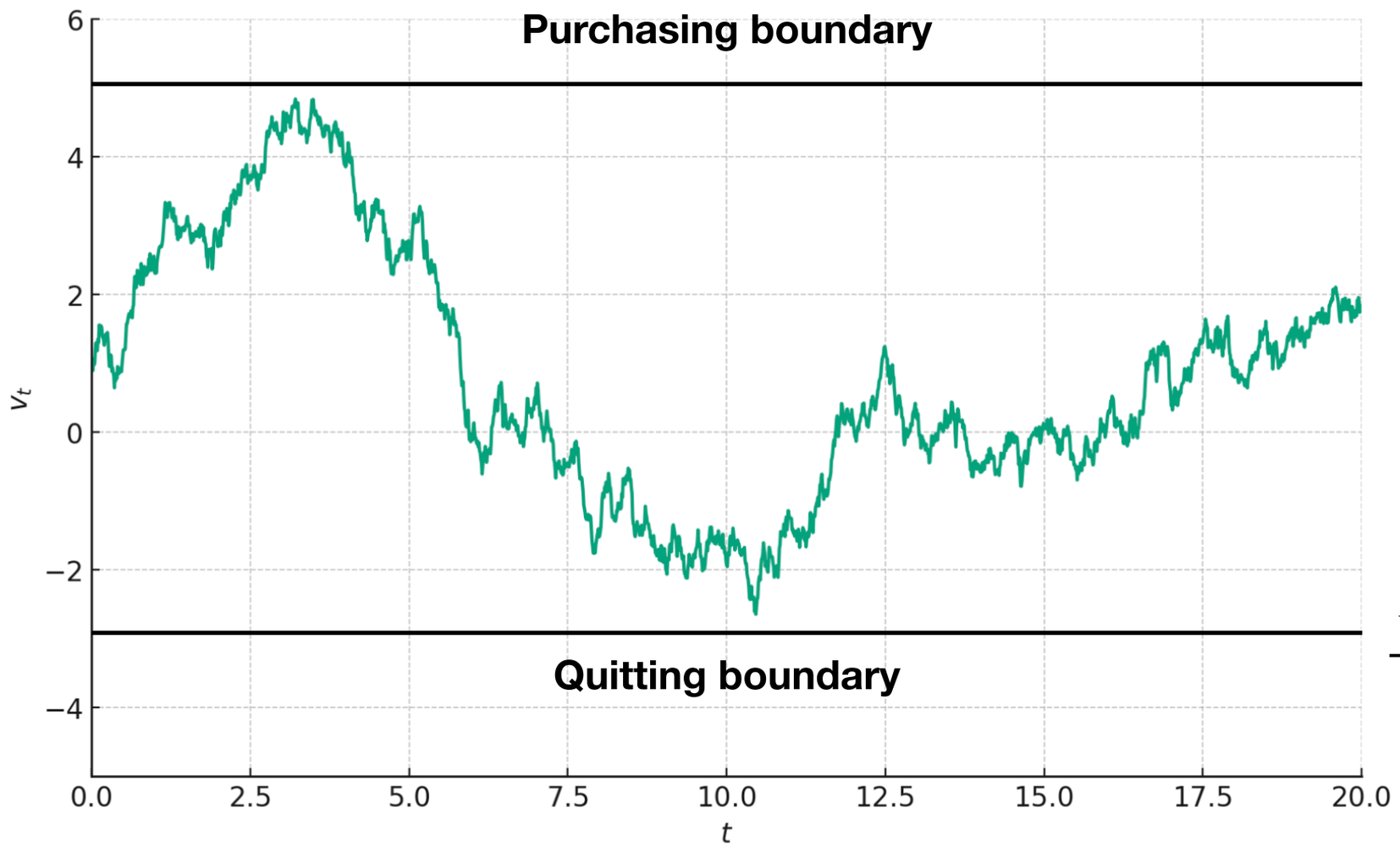
  consumer's valuation: $dv_t = \sigma dW_t$ (a Brownian motion)
- No discounting

Sample path of the consumer's learning processes



$$v_0 = 1$$

Consumer's search strategy



$\bar{V}_1(p) = \sigma^2/4c + p$

$\underline{V}_1(p) = -\sigma^2/4c + p$

$v_0 = 1,\ p = 1$

Consumer search can benefit the firm

Without consumer search: market does not exist

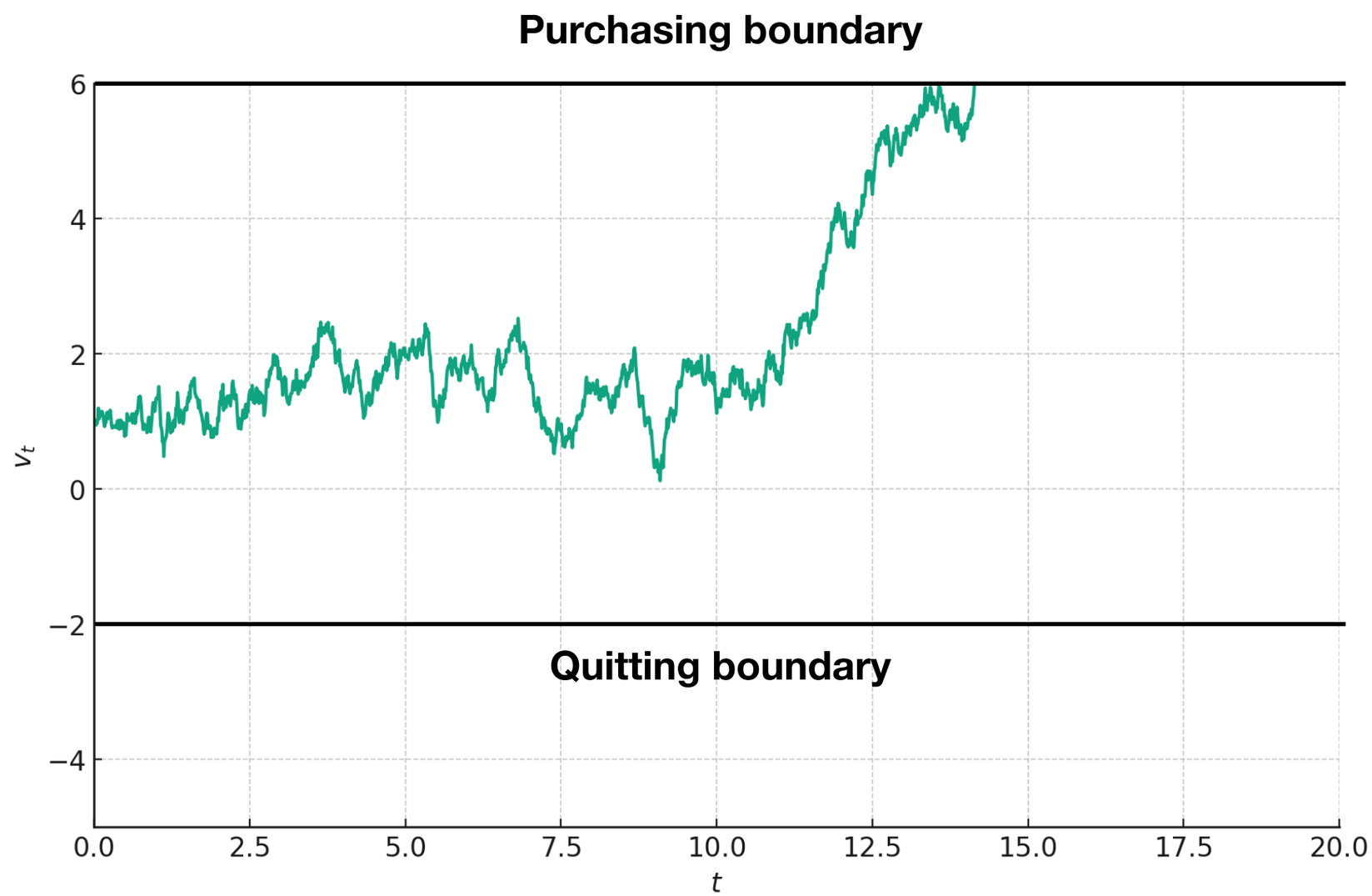
- $v_0 = 1 < m = 1.5$:

Consumer will buy if $p \leq 1$

Firm will sell if $p \geq 1.5$

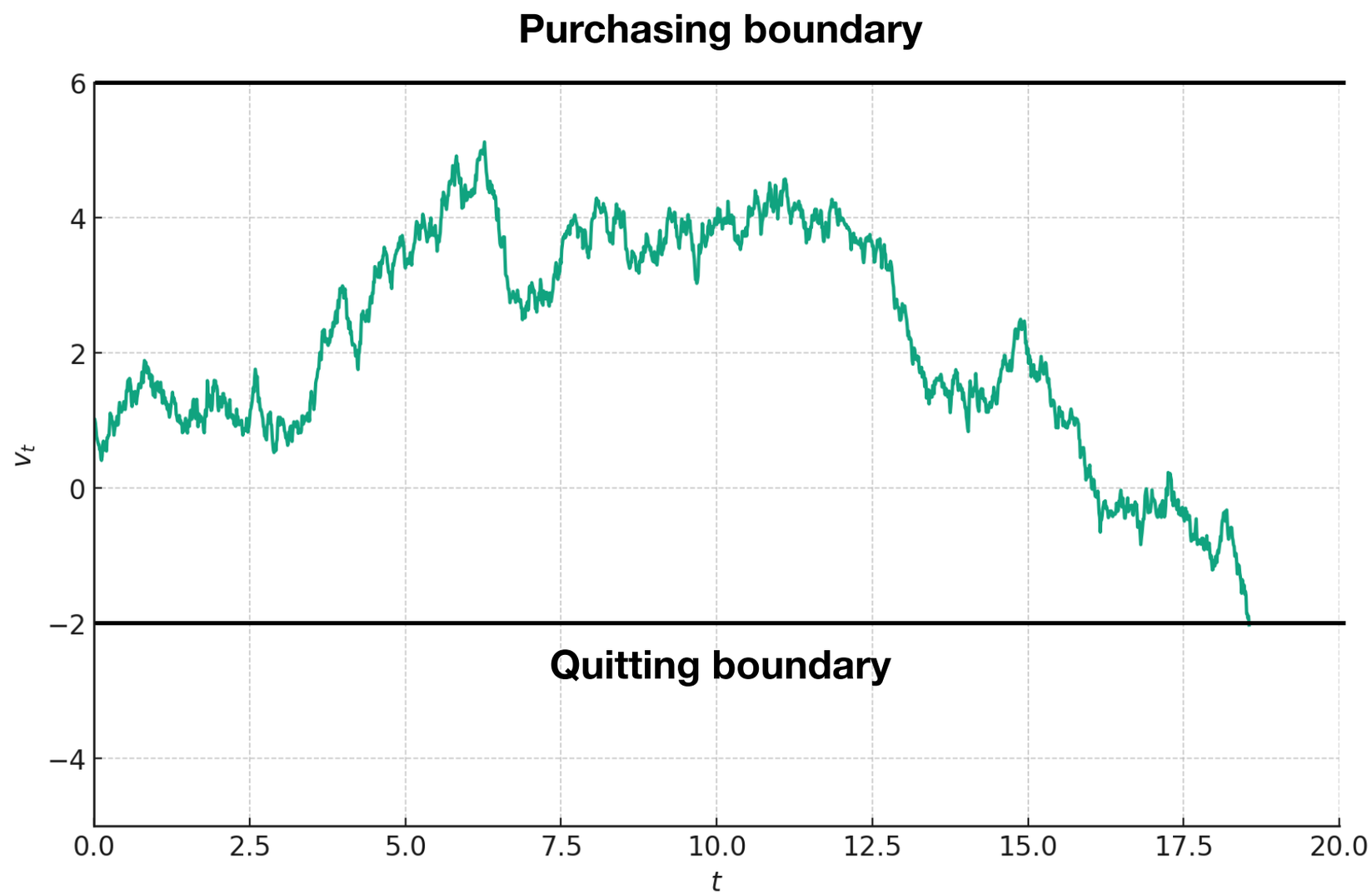
$\Rightarrow$ No market

With consumer search: purchase



$v_0 = 1,\ m = 1.5,\ p = 2$

With consumer search: no purchase



$v_0 = 1,\ m = 1.5,\ p = 2$

Consumer search can benefit the firm

- When search is not feasible, the firm can charge at most v_0

The firm will not sell any product if the initial valuation is lower than the marginal cost, $v_0 < m$

- When search is feasible, the firm can charge a higher price

The firm can make positive profits even if $v_0 < m$

Equilibrium without a hidden fee (with consumer search)

Without a hidden fee (with consumer search)

- For a given valuation v and upfront price p_{wo}, the consumer's purchasing probability:

$$Q_1(v, p_{wo}) = \begin{cases} 1, & \text{if } p_{wo} \leq v - \frac{\sigma^2}{4c}, \\ \frac{v + \frac{\sigma^2}{4c} - p_{wo}}{\frac{\sigma^2}{2c}}, & \text{if } v - \frac{\sigma^2}{4c} < p_{wo} < v + \frac{\sigma^2}{4c}, \\ 0, & \text{if } p_{wo} \geq v + \frac{\sigma^2}{4c}, \end{cases}$$

- Expected profit: $(p^*_{wo} - m) \cdot Q_1(v_0, p_{wo})$

Without a hidden fee (with consumer search)

- Optimal price:

$$p^*_{wo} = \begin{cases} v_0 - \frac{\sigma^2}{4c},\ if\ v_0 \geq \frac{3\sigma^2}{4c} + m \\ \frac{v_0}{2} + \frac{\sigma^2}{8c} + \frac{m}{2},\ if\ -\frac{\sigma^2}{4c} + m < v_0 < \frac{3\sigma^2}{4c} + m \end{cases}$$

- Ex-ante purchasing probability given the optimal price:

$$Q_1(v_0, p^*_{wo}) \begin{cases} 1,\ if\ v_0 \geq \frac{3\sigma^2}{4c} + m, \\ \frac{c}{\sigma^2}(v_0 + \sigma^2/4c - m),\ if\ -\frac{\sigma^2}{4c} + m < v_0 < \frac{3\sigma^2}{4c} + m, \\ 0,\ if\ v_0 \leq -\frac{\sigma^2}{4c} + m \end{cases}$$

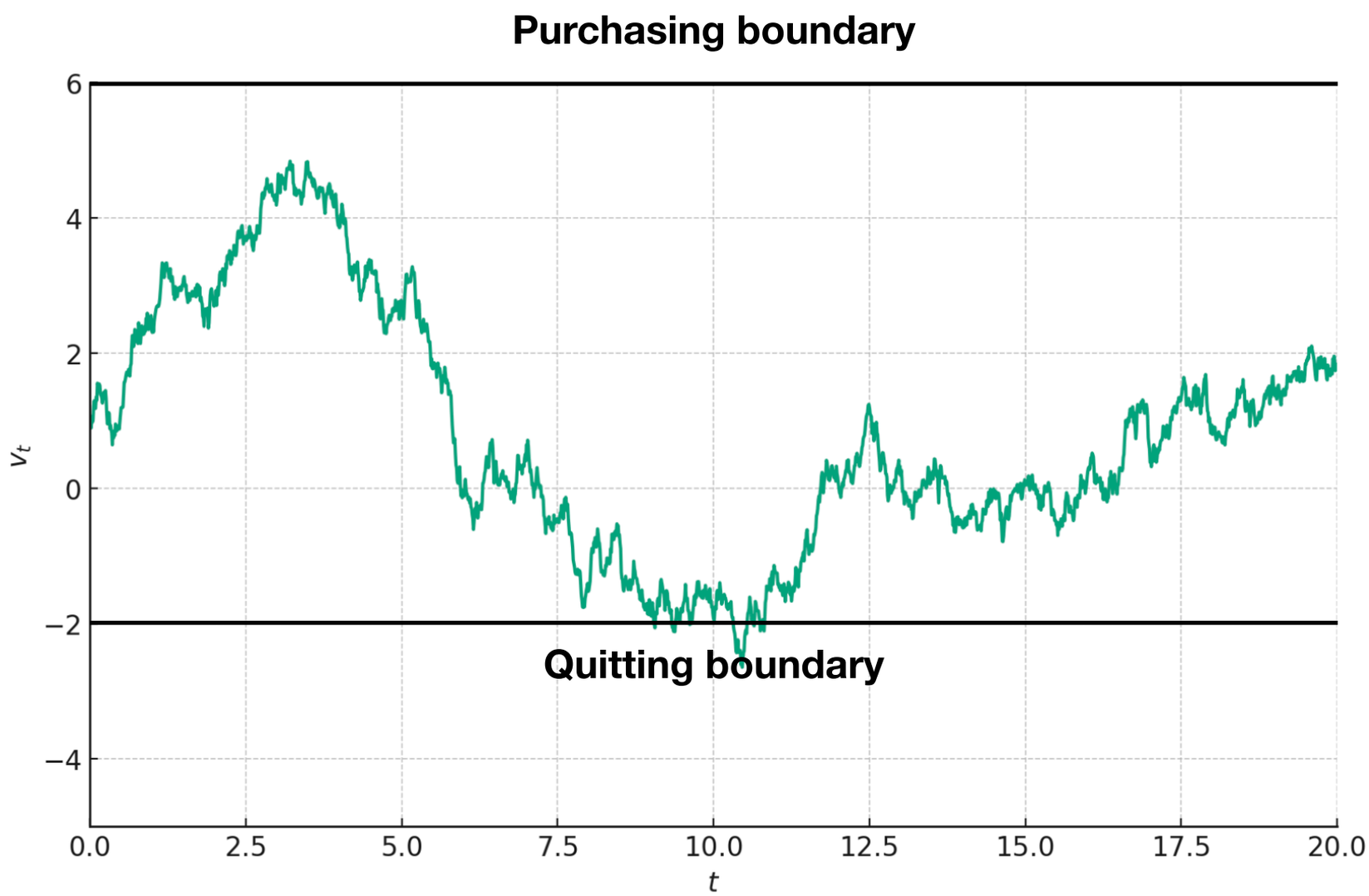
Expected search time and consumer welfare

- $v_0 \leq -\frac{\sigma^2}{4c} + m$ (quit directly): 0 search time τ_{wo}, 0 consumer welfare
- $v_0 \geq \dfrac{3\sigma^2}{4c} + m$ (buy directly): 0 search time, consumer welfare is $v_0 - p^*_{wo} = \sigma^2/4c$
- $-\frac{\sigma^2}{4c} + m < v_0 < \frac{3\sigma^2}{4c} + m$ (search): Dynkin's formula $\Rightarrow$

$$\mathbf{E}(\tau_{wo}) = \frac{\frac{\sigma^4}{16c^2} - (v_0 - p^*_{wo})^2}{\sigma^2}$$ (maximized when $p^*_{wo} = v_0$)

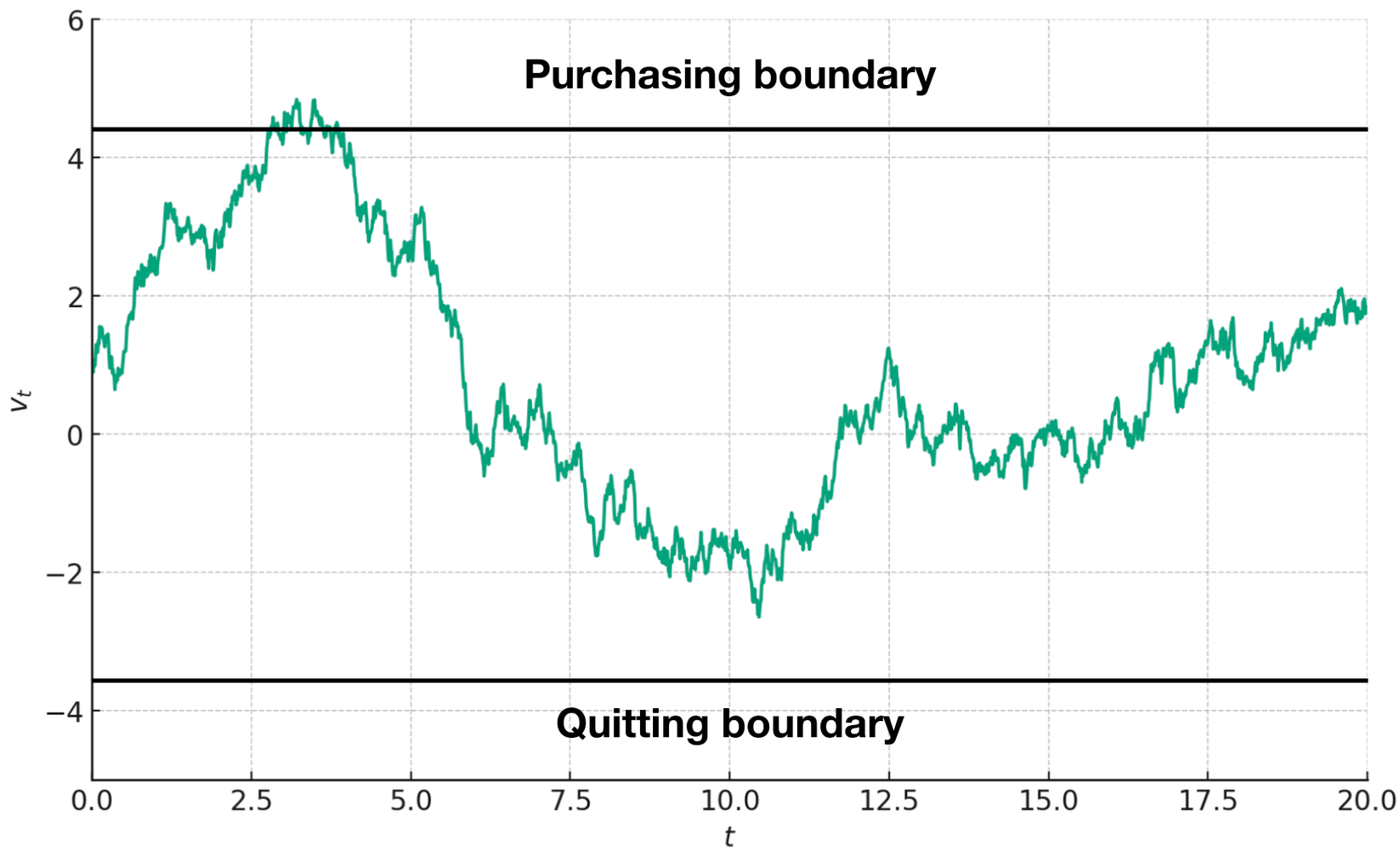
$$\mathbf{E}(consumer\ surplus) = -\,c \cdot \mathbf{E}(\tau_{wo}) + \frac{\sigma^2}{4c} \cdot Q_1(v_0, p^*_{wo})$$

Expected search time



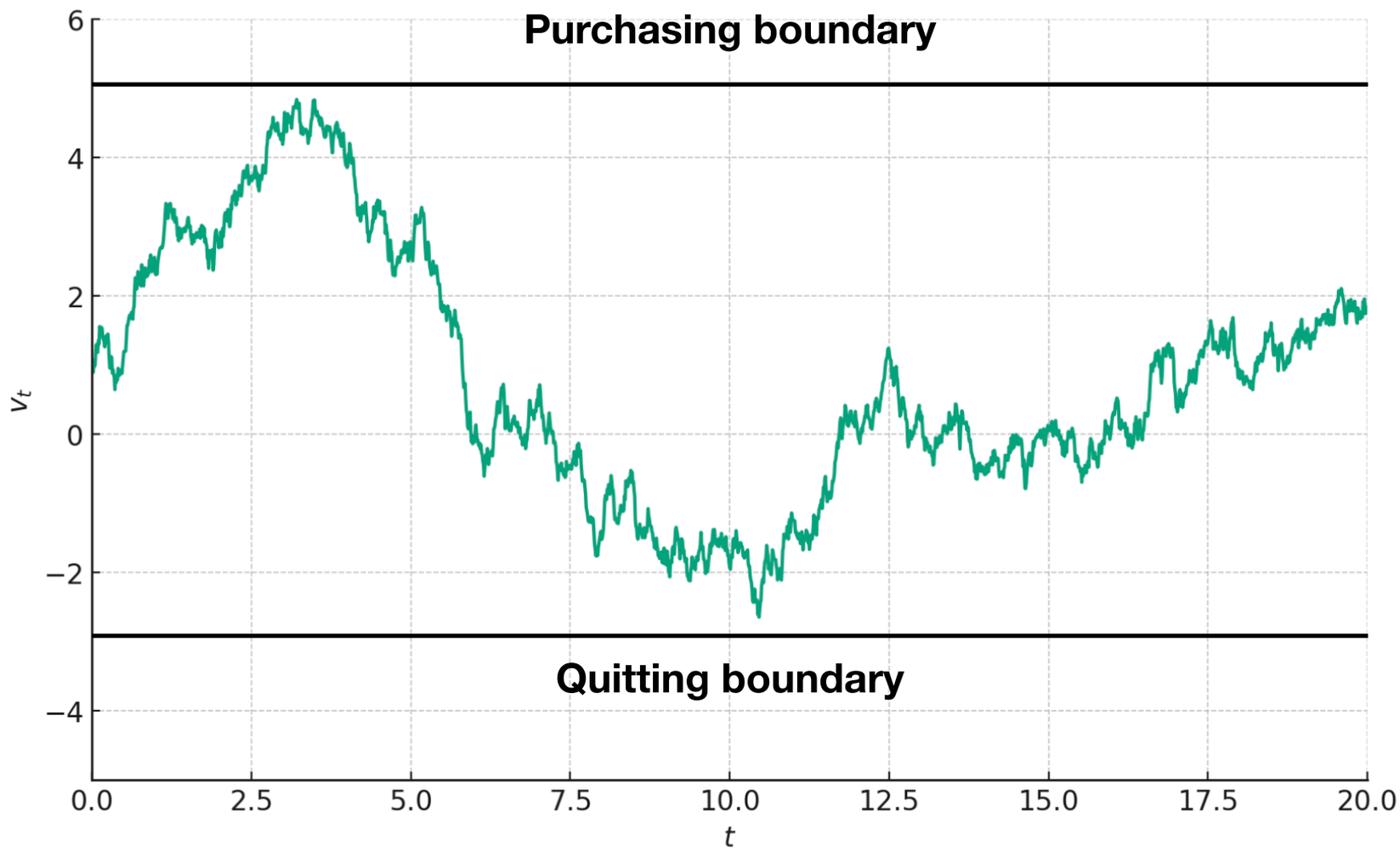
$v_0 = 1,\ p = 2$

Expected search time



$v_0 = 1,\ p = 0.5$

Expected search time



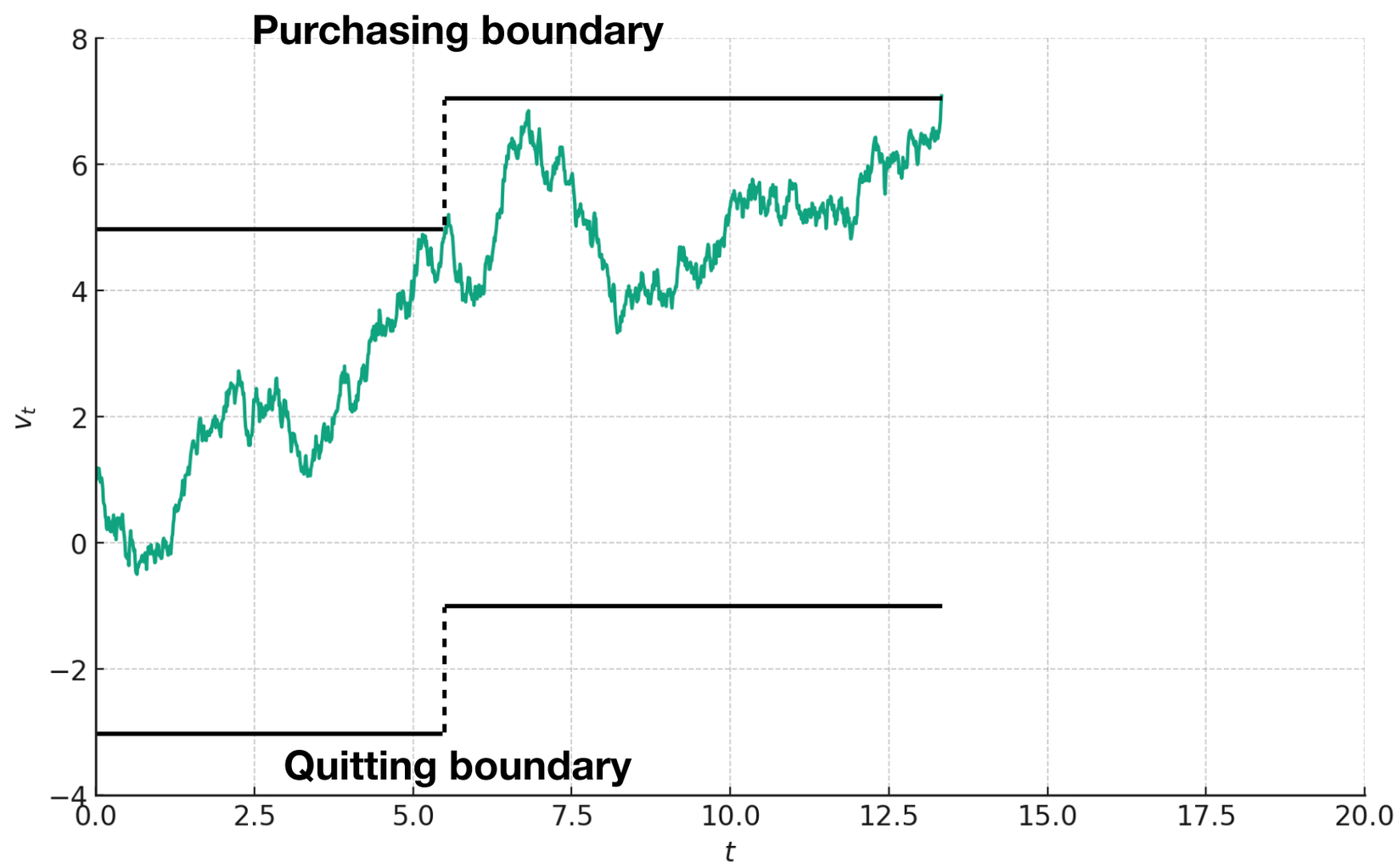
$v_0 = 1,\ p = 1$

Equilibrium with a hidden fee (with consumer search)

+ Hidden fee

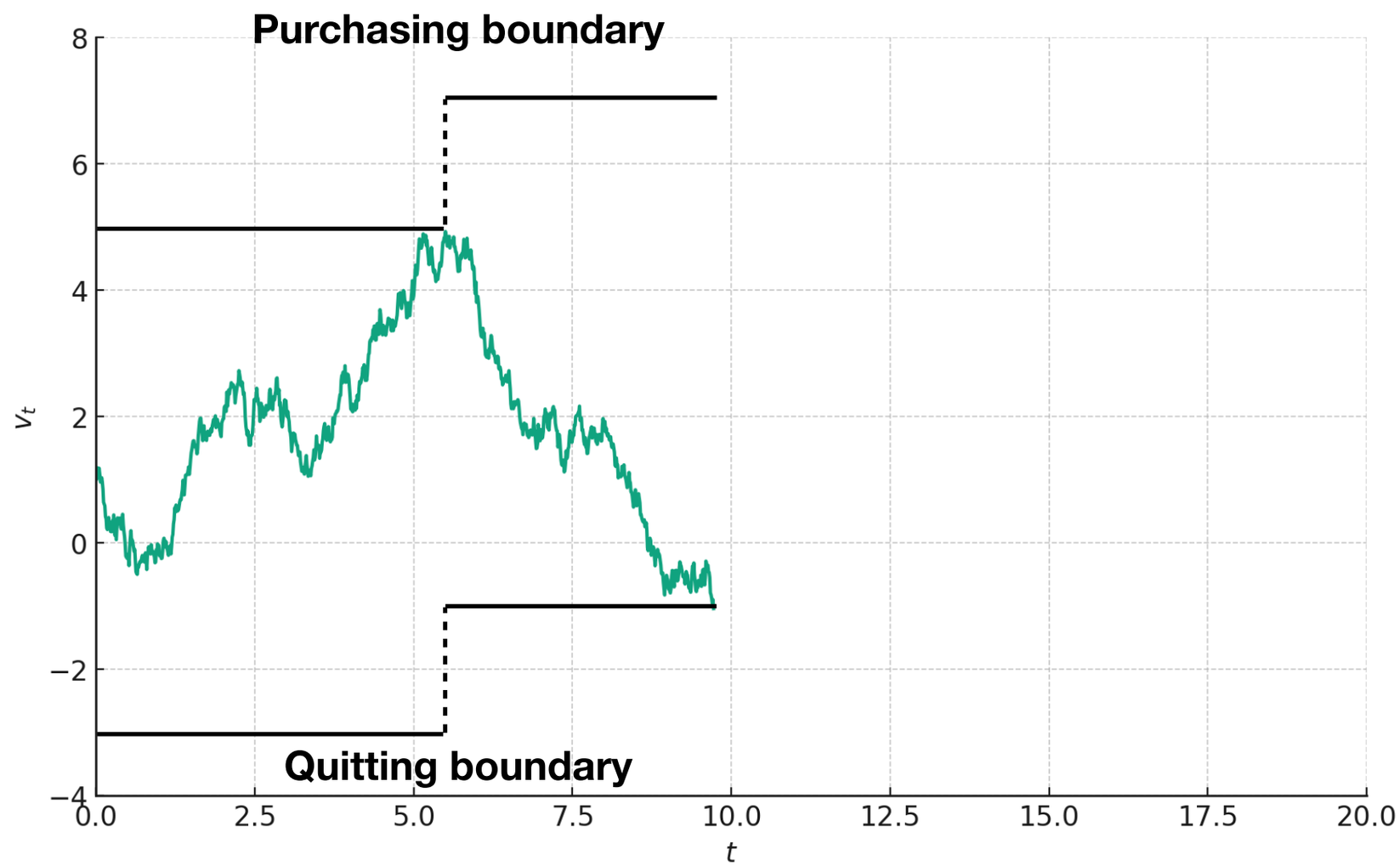
- The same setup as before, except that:
- When a consumer's valuation v_t reaches the purchasing threshold, she
  - decides to buy and go to the checkout page
  - will see an additional hidden fee Δp
  - will not buy the product (the hidden fee raises the purchasing threshold)
  - faces an updated search problem

Sample search path (purchase)



$v_0 = 1,\ m = 1.5,\ p = 1,\ \Delta p = 2$

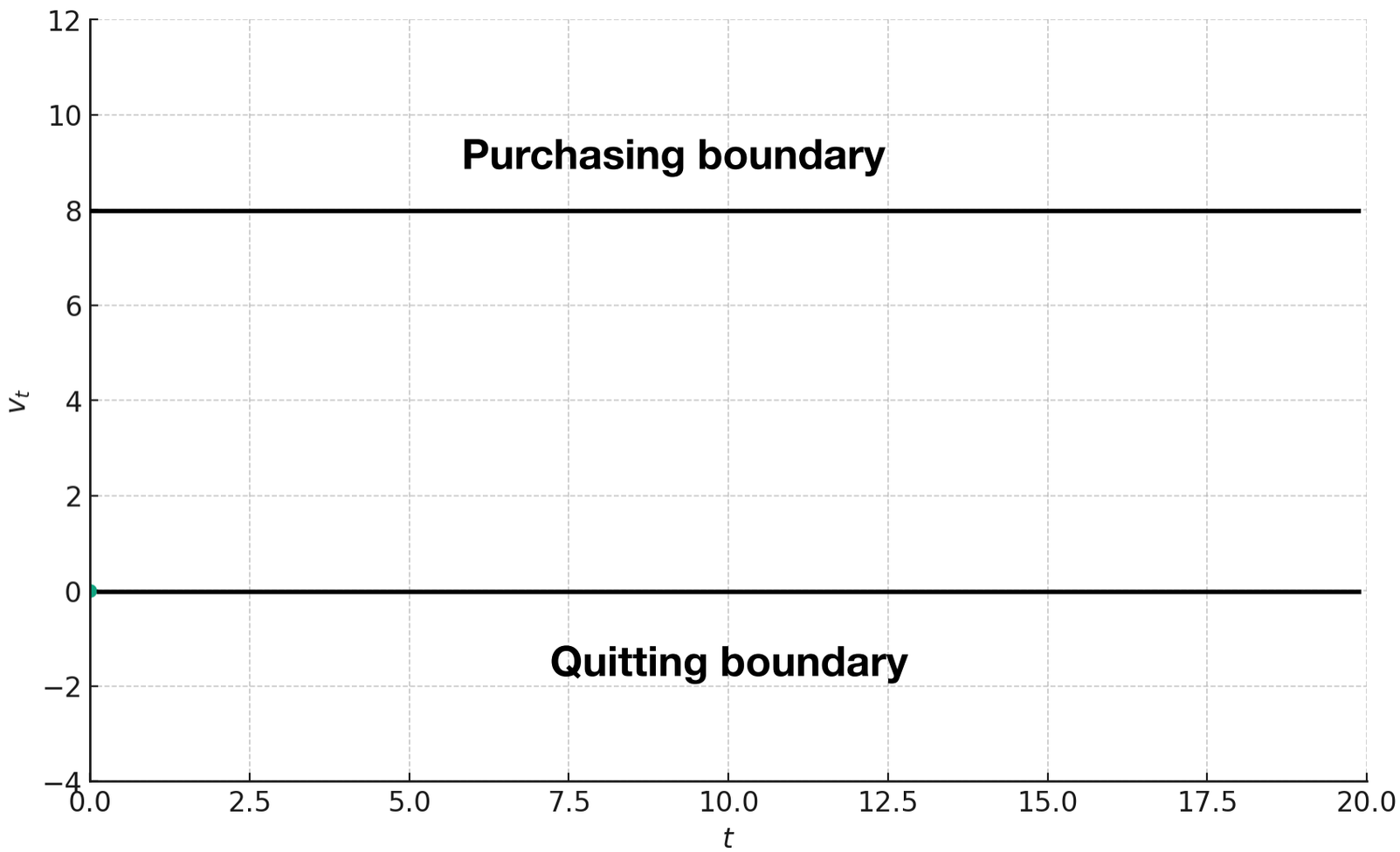
Sample search path (no purchase)



$v_0 = 1,\ m = 1.5,\ p = 1,\ \Delta p = 2$

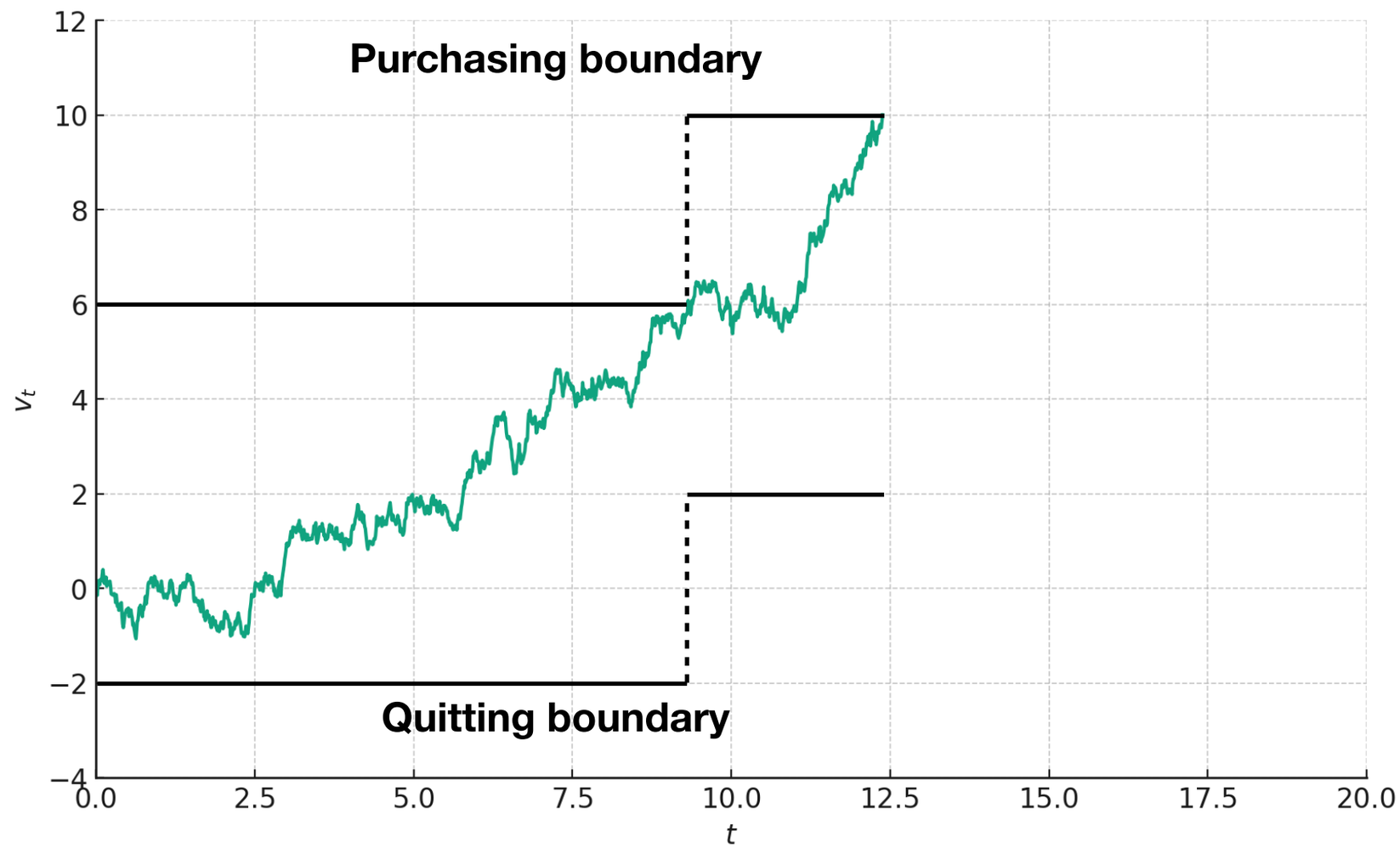
Hidden fees can benefit the firm

Neither search nor purchase without a hidden fee



$v_0 = 0,\ m = 4,\ p = 4$

Positive expected profit with a hidden fee



$v_0 = 0,\ m = 4,\ p_1 = 2,\ \Delta p = 4$

Hidden fees can benefit the firm

- When hidden fee is not feasible, the consumer may search or buy directly only if $v_0 > p - \sigma^2/4c$

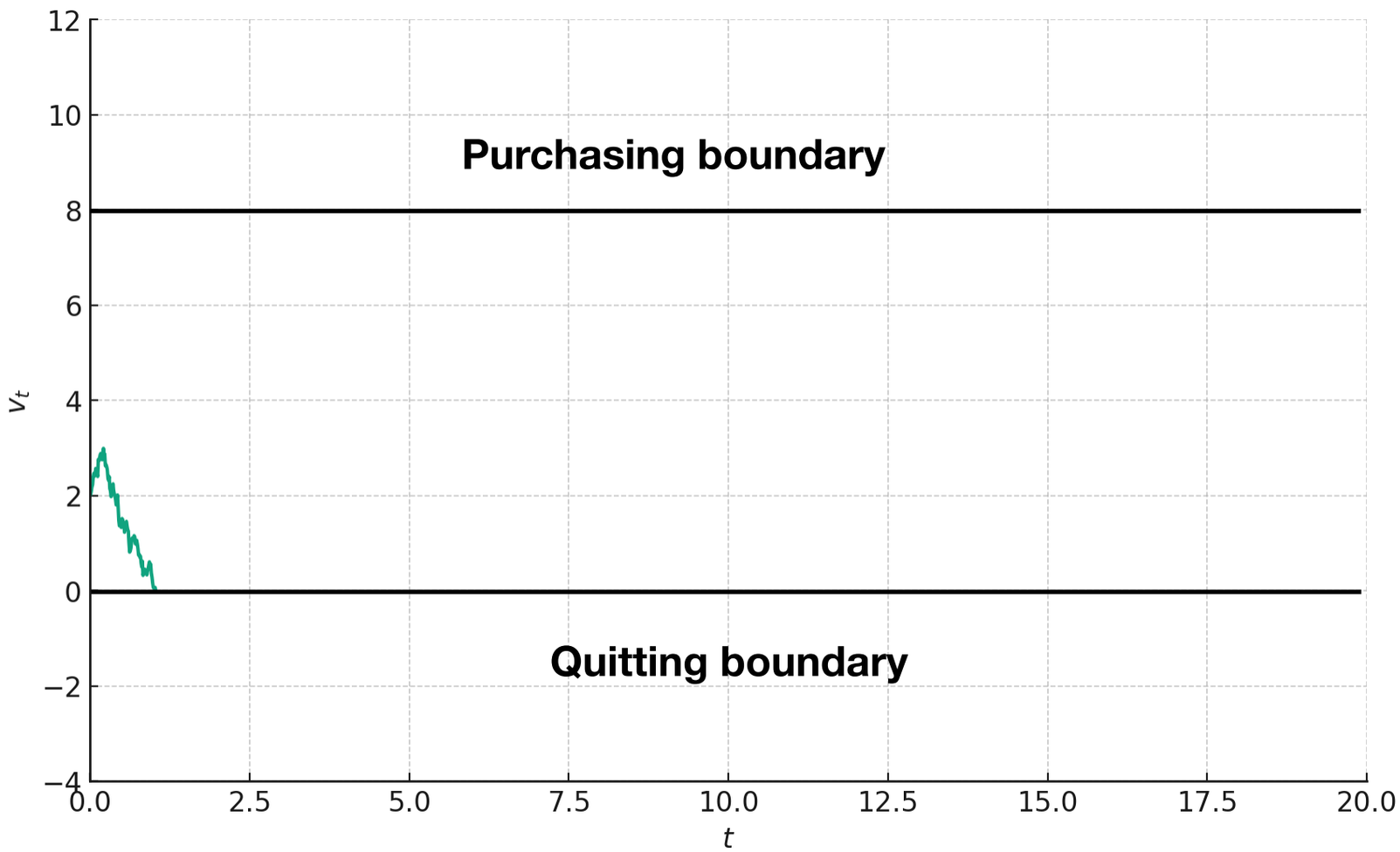
- To earn a profit, price must be higher than marginal cost p > m

The firm will not sell any product if $v_0 < -\sigma^2/4c + m$

- When hidden fee is feasible, the firm can induce the consumer to search in a wider range by offering a lower price initially

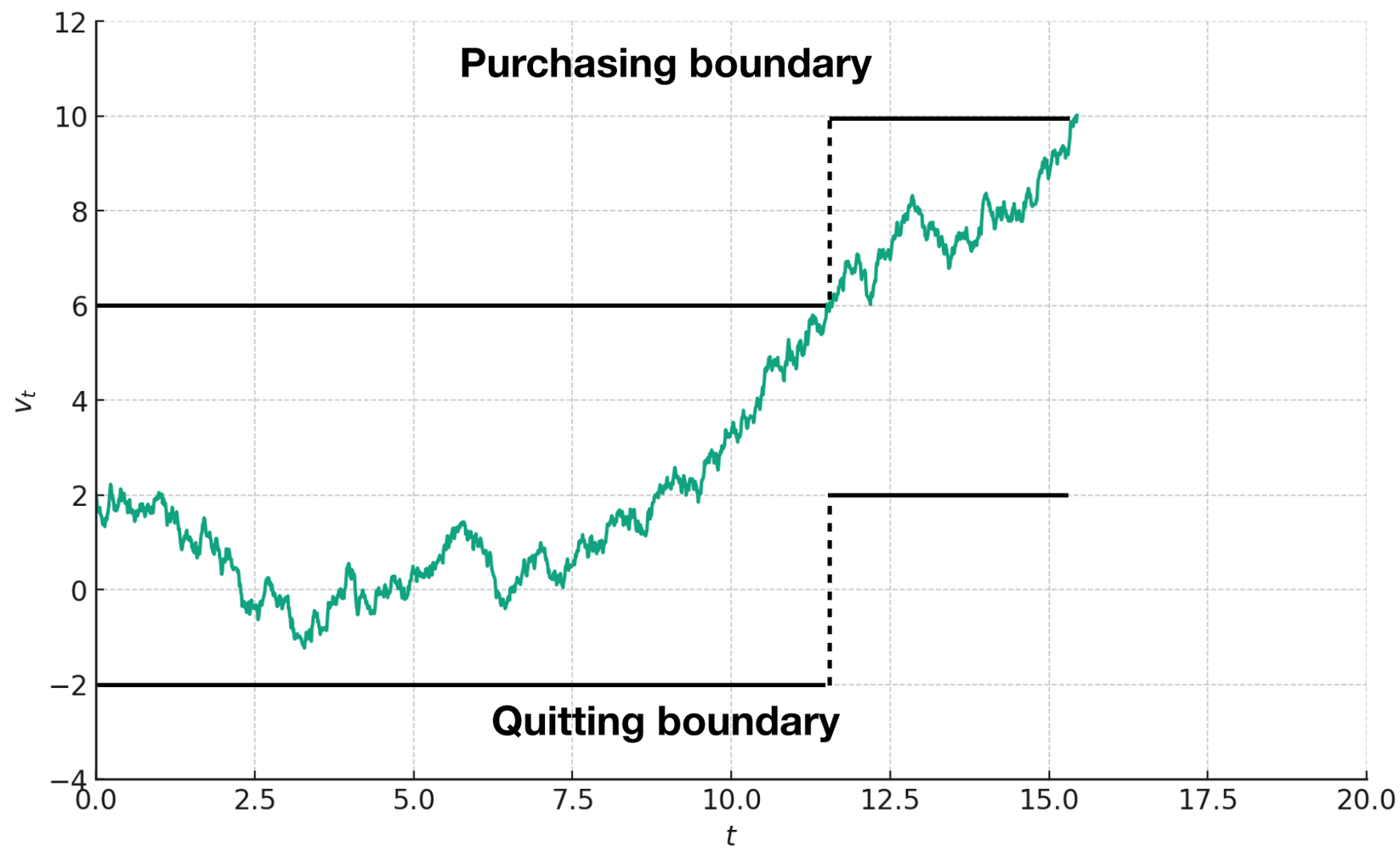
The firm can make positive profits even if $v_0 < -\sigma^2/4c + m$

Quick exit after searching without a hidden fee



$v_0 = 2,\ m = 2,\ p = 4$

Purchase after searching with a hidden fee



$v_0 = 2,\ m = 2,\ p_1 = 2,\ \Delta p = 4$

Hidden fees can benefit the firm

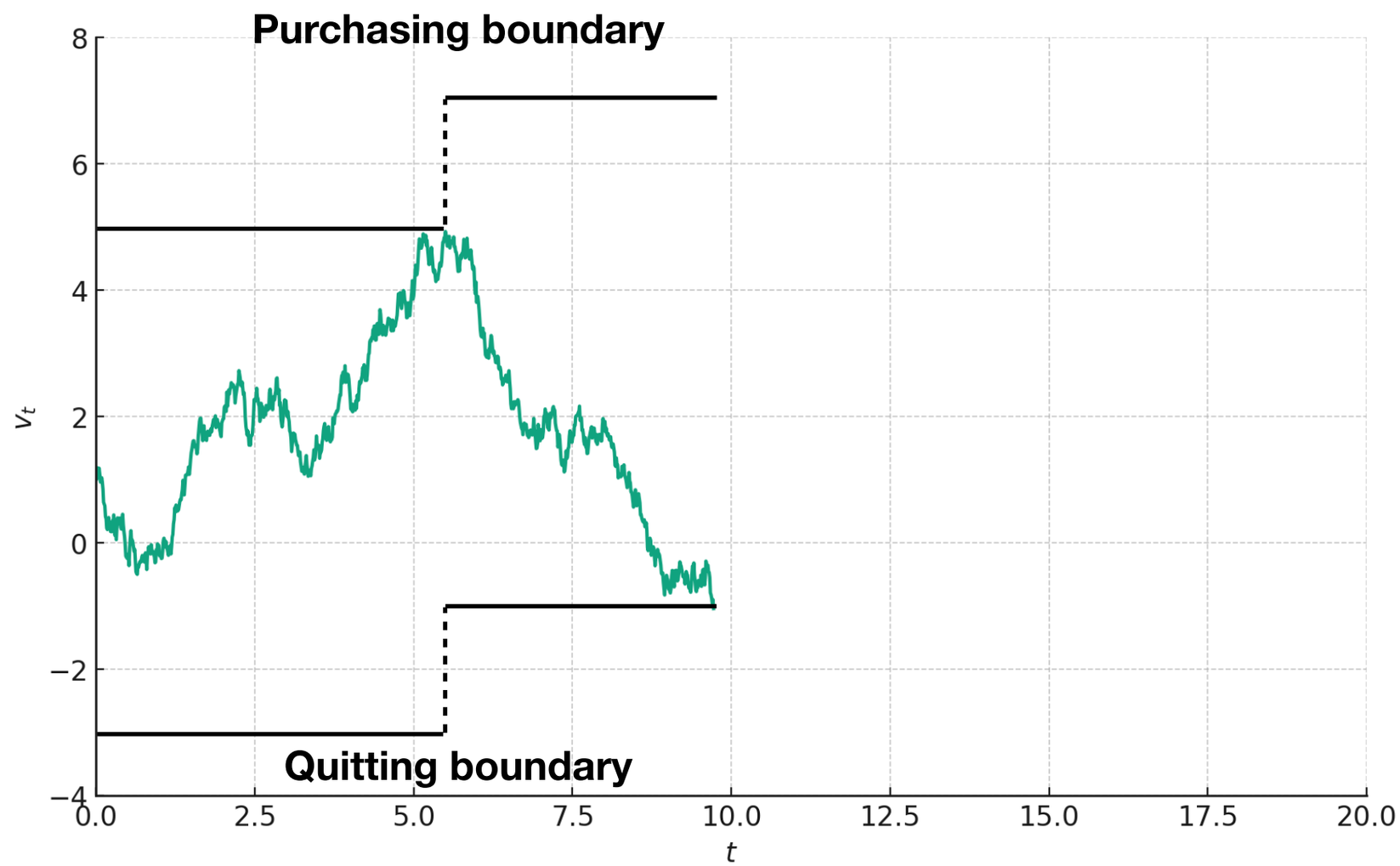
- Even if the firm can earn a positive profit without hidden fee when $v_0 > -\sigma^2/4c + m$, it can increase the profit by using hidden fee

$\Rightarrow$ induce consumers to continue searching in cases where early signals are unfavorable (more persistent in search)

- Parameter range: $-\sigma^2/4c + m < v_0 < 3\sigma^2/4c + m$

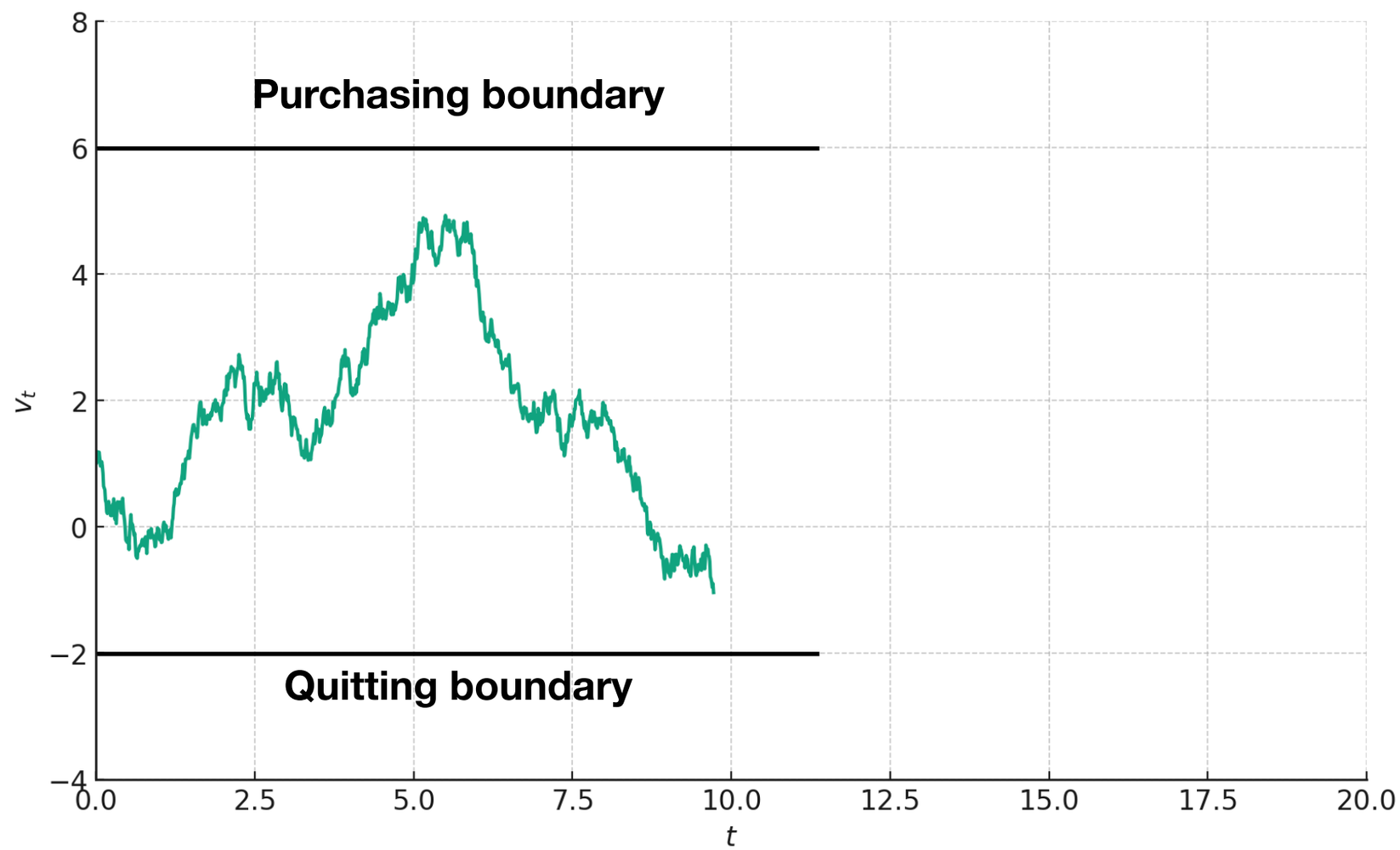
Hidden fees can hurt the firm

Quit under hidden fee



$v_0 = 1,\ m = 1.5,\ p = 1,\ \Delta p = 2$

Still searching under upfront price



$v_0 = 1,\ m = 1.5,\ p = 2$

Optimal price with a hidden fee

Optimal price with a hidden fee

- When consumer valuation reaches the purchasing threshold $\bar{V}_1(p_1)$, the consumer decides to buy and proceeds to the checkout page

- The hidden fee $\Delta p > 0$ is revealed, increasing the purchasing threshold to $\bar{V}_1(p_1 + \Delta p) = \bar{V}_1(p_1) + \Delta p$

- If the initial price p_1 induces immediate purchase/quit $\Rightarrow$ the same as the upfront pricing case

- Focus on the case where p_1 induces search, $p_1 \in (v_0 - \sigma^2/4c, v_0 + \sigma^2/4c)$

Optimal price with a hidden fee

- For a given p_1, the probability that the consumer goes to the checkout page (valuation reaches $\bar{V}_1(p_1)$ before hitting $\underline{V}_1(p_1)$):

$$Q_1(v_0, p_1) = \frac{v_0 + \sigma^2/4c - p_1}{\sigma^2/2c}$$

- If the hidden fee Δp is too high, the consumer will quit immediately

$\Rightarrow$ in equilibrium, the hidden fee must be moderate, $\Delta p \in (0, \sigma^2/2c)$

- Purchasing probability conditional on reaching the checkout page:

$$Q_2(\Delta p) = 1 - \frac{\Delta p}{\sigma^2/2c}$$

Optimal price with a hidden fee

- Firm's overall expected profit:

$$
\begin{aligned}
&\Pi_w(p_1, \Delta p) \\
&= \underbrace{(p_1 + \Delta p - m)}_{\text{profit per sale}} \cdot \underbrace{Q_1(v_0, p_1)}_{\text{probability of reaching the checkout page}} \cdot \underbrace{Q_2(\Delta p)}_{\text{conditional probability of purchasing}} \\
&= (p_1 + \Delta p - m) \cdot \frac{2c}{\sigma^2}(v_0 + \frac{\sigma^2}{4c} - p_1) \cdot (1 - \frac{2c}{\sigma^2}\Delta p)\,.
\end{aligned}
$$

Optimal price with a hidden fee

- Firm's constrained optimization problem:

$$\max_{p_1,\Delta p} \; (p_1 + \Delta p - m) \cdot \frac{2c}{\sigma^2}(v_0 + \frac{\sigma^2}{4c} - p_1) \cdot (1 - \frac{2c}{\sigma^2}\Delta p)$$

$$s.t. \;\; p_1 \geq 0,$$

$$p_1 \in (v_0 - \sigma^2/4c, v_0 + \sigma^2/4c),$$

$$\Delta p \in [0, \sigma^2/2c).$$

Expected search time induced by the optimal price

- Relatively high initial valuation v_0

$$\mathbf{E}(\tau_1) = \mathbf{E}(\tau_2)$$

The firm **perfectly smooths** consumers' search behavior across stages

- Intuition: (consider a fixed total price)

Initial price too high ⇒ quick exit if gathering a few negative signals early on ❌

Initial price too low 1) ⇒ go to the checkout stage with a high probability

2) ⇒ a high hidden fee ⇒ quick exit ❌

The optimal price balances consumers' incentives to continue search before and after fee disclosure

Expected search time induced by the optimal price

- Relatively low initial valuation v_0

$$\mathbf{E}(\tau_1) < \mathbf{E}(\tau_2)$$

- Intuition:

Low $v_0 \Rightarrow$ quitting boundary is close to v_0 even with a low initial price
$\Rightarrow$ quick exit if the consumer gathers a few negative signals

If a consumer reaches the checkout page $\Rightarrow$ a much higher valuation
$\Rightarrow$ willing to keep searching even if she receives some negative signals

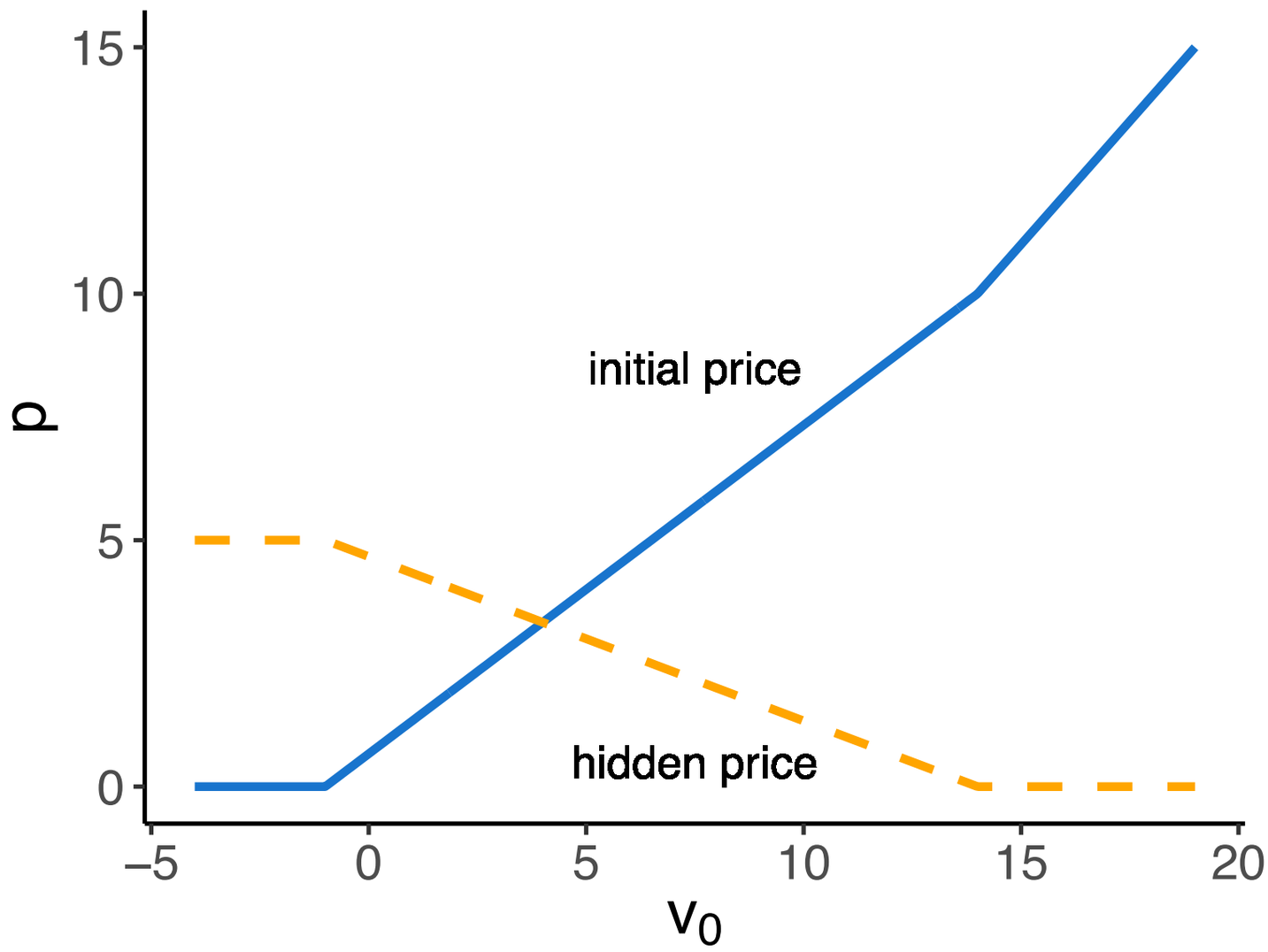
Impossible for the firm to perfectly smooth consumers' search behavior.

Optimal price with a hidden fee

Proposition 2. *If* $\max\{-m/2, -3\sigma^2/4c + m\} < v_0 < 3\sigma^2/4c + m$*, then the optimal initial price is* $p_1^* = 2v_0/3 + m/3$ *and the optimal hidden fee is* $\Delta p^* = \sigma^2/4c + (m - v_0)/3$*. The optimal total price is* $p_1^* + \Delta p^* = v_0/3 + \sigma^2/4c + 2m/3$*, which is strictly higher than the optimal price without a hidden fee,* p_{wo}^**.*

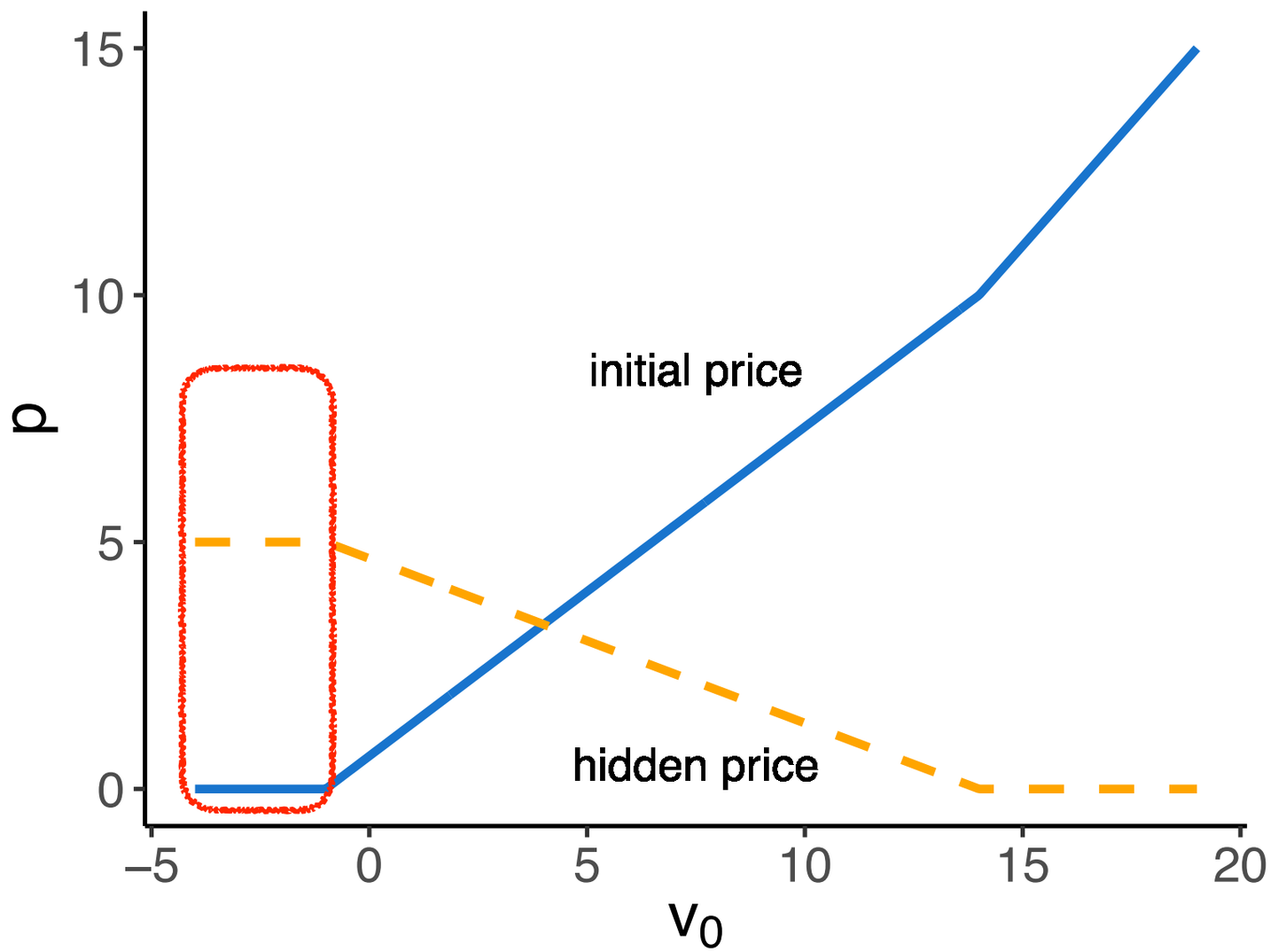
If $-\sigma^2/4c < v_0 \leq -m/2$*, then the optimal initial price is* $p_1^* = 0$ *and the optimal hidden fee is* $\Delta p^* = \sigma^2/4c + m/2$*.*

Optimal Price with a hidden fee



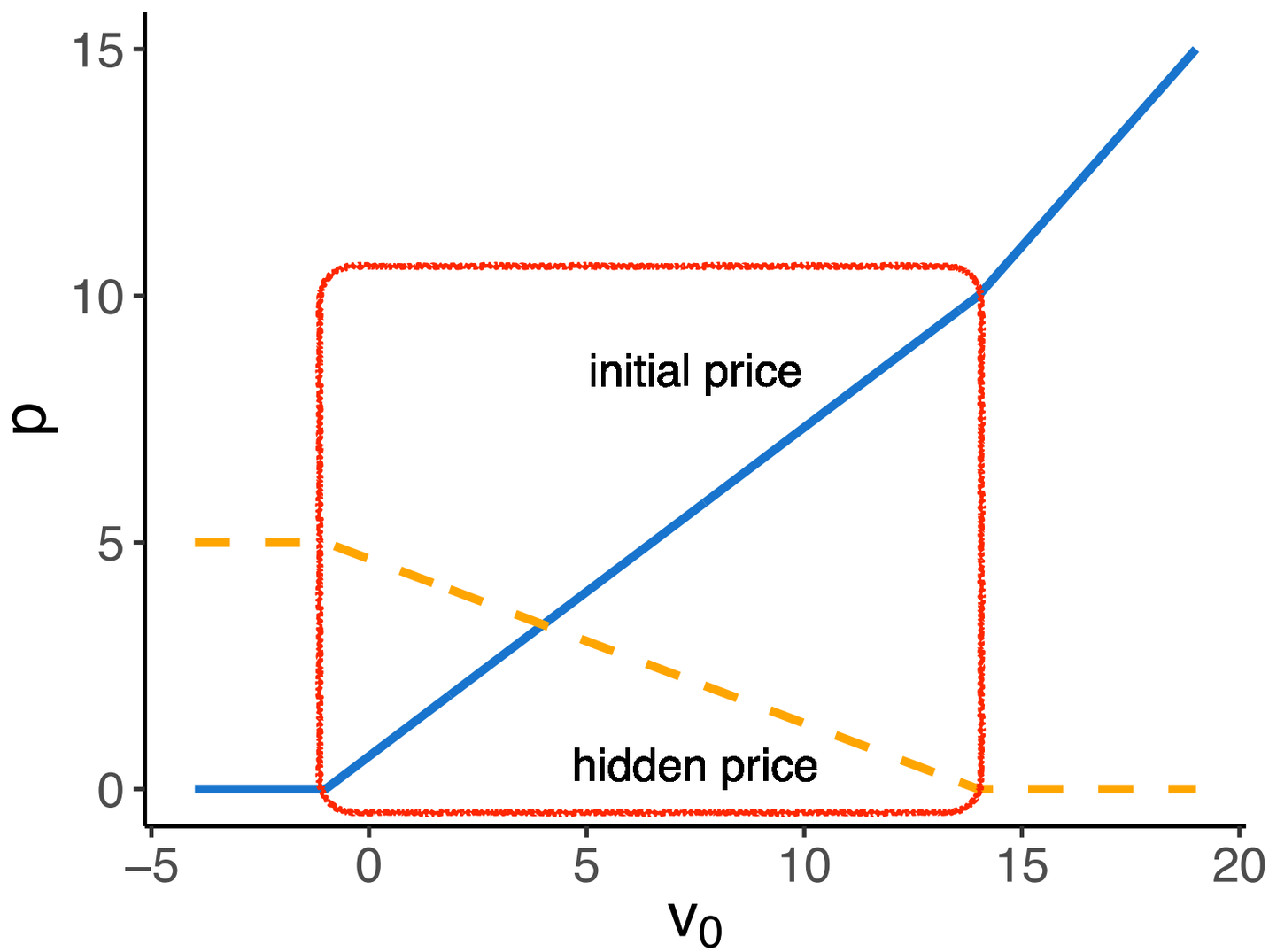
$m = 2,\ \sigma = 0.4,\ c = 0.01$

Optimal Price with a hidden fee



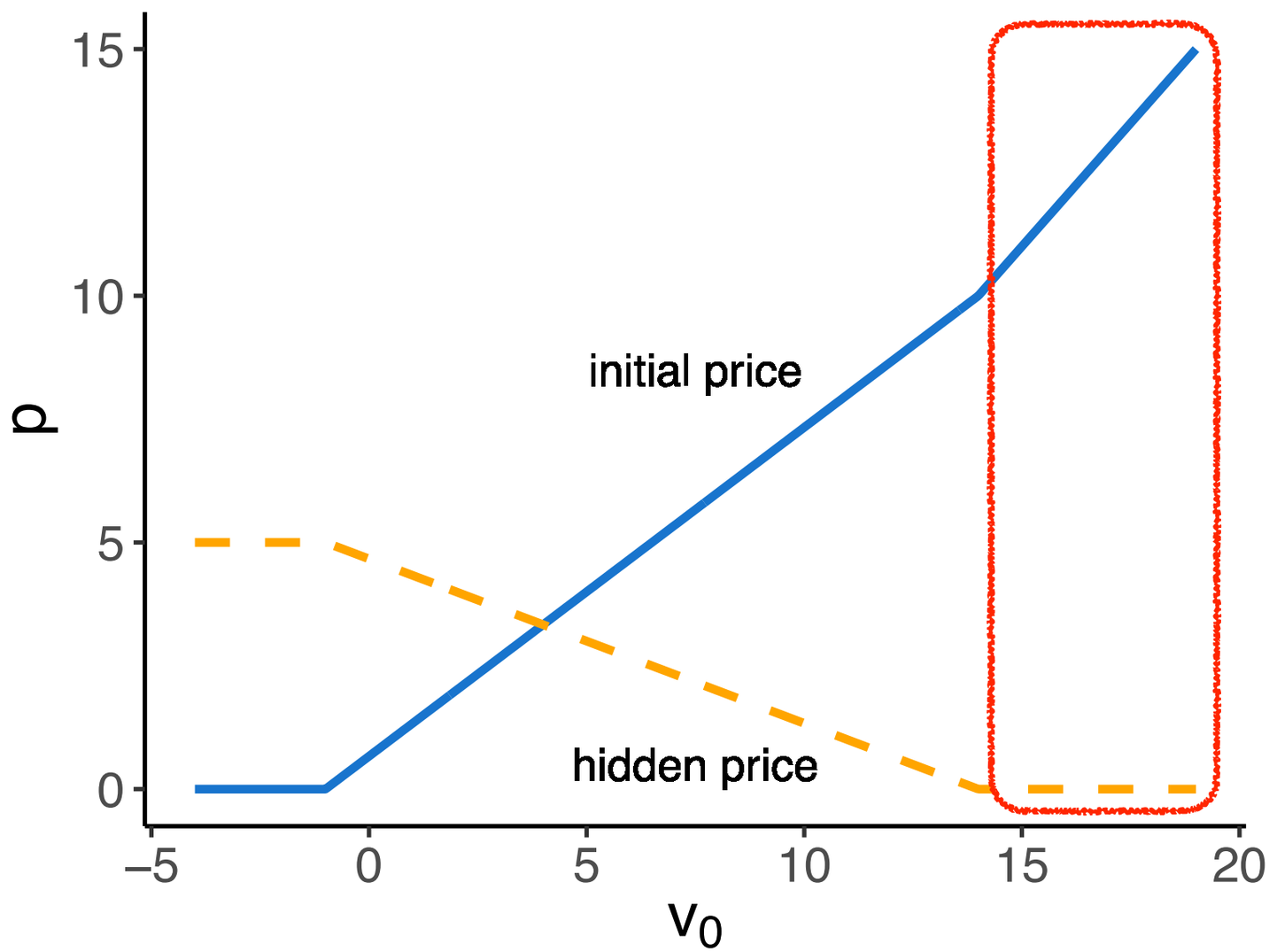
$m = 2,\ \sigma = 0.4,\ c = 0.01$

Optimal Price with a hidden fee



$m = 2,\ \sigma = 0.4,\ c = 0.01$

Optimal Price with a hidden fee



$m = 2,\ \sigma = 0.4,\ c = 0.01$

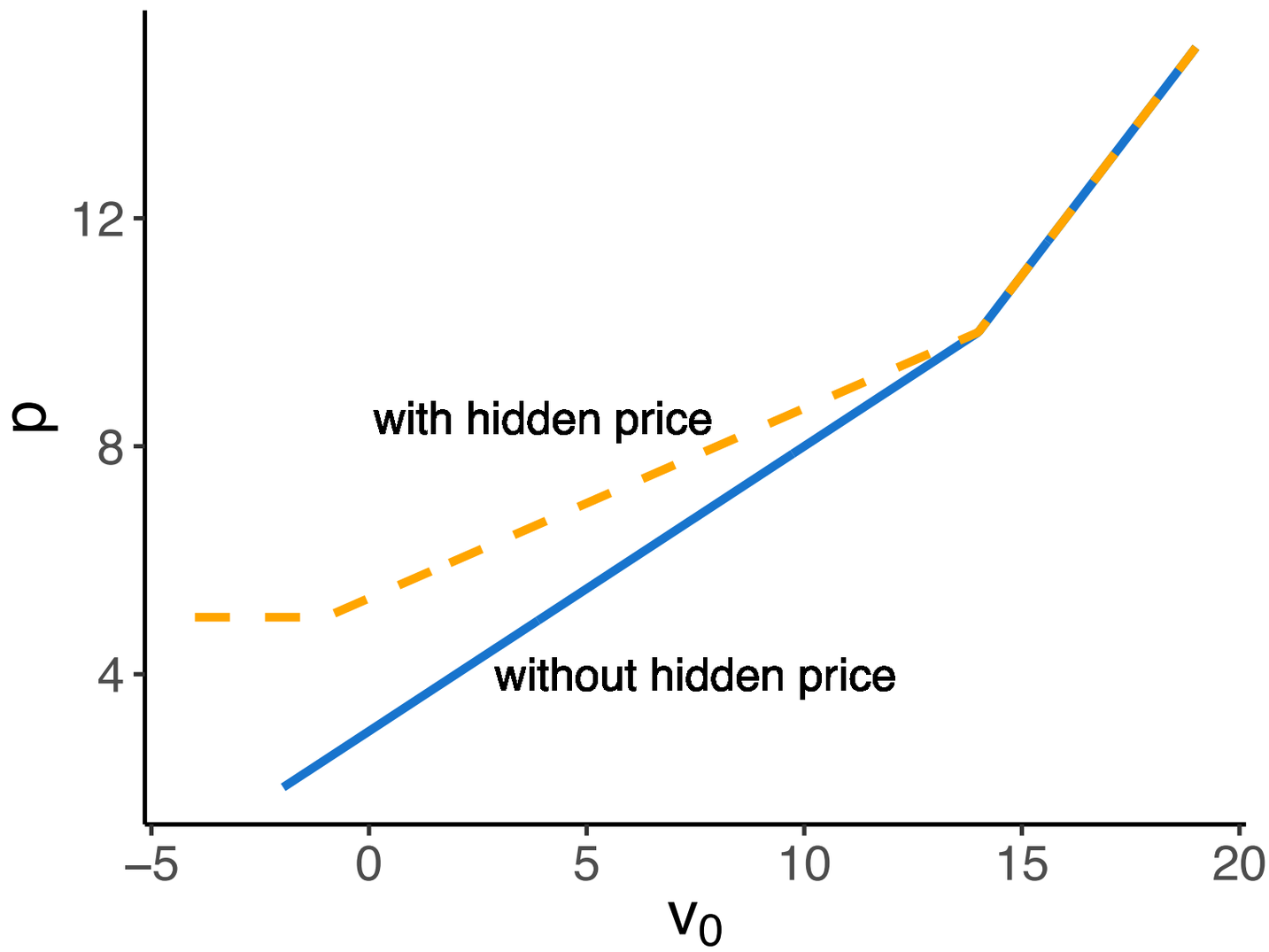
Comparative statics of the optimal price with hidden fee

- The hidden fee Δp increases in the signal informativeness σ^2, the marginal cost m, and decreases in the search cost c, the initial valuation v_0

- The initial price p increases in v_0 and m (it does not depend on σ^2 and c)

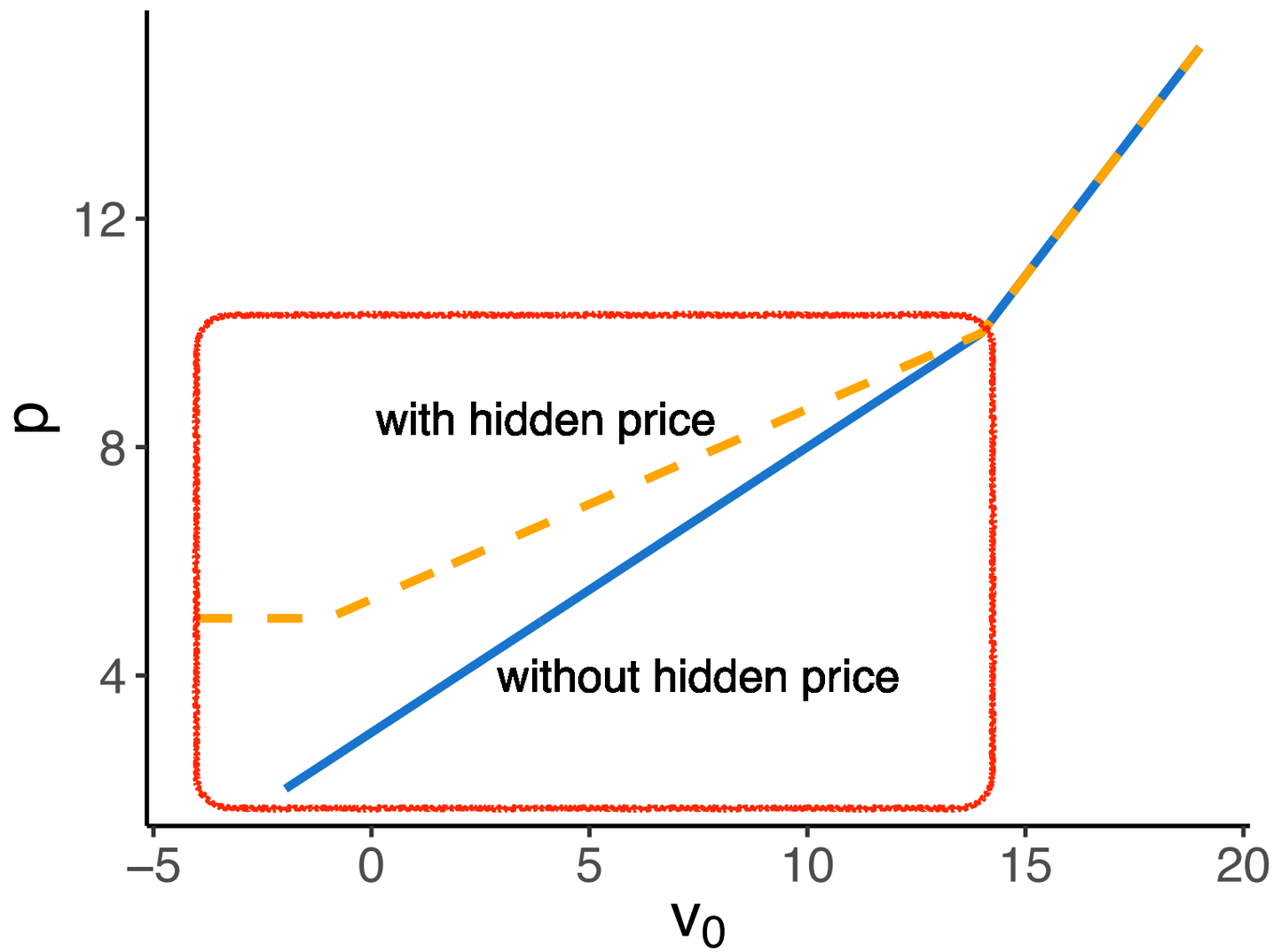
- The total price $p + \Delta p$ increases in v_0, σ^2, m, and decreases in c

Optimal (total) price with and without a hidden fee

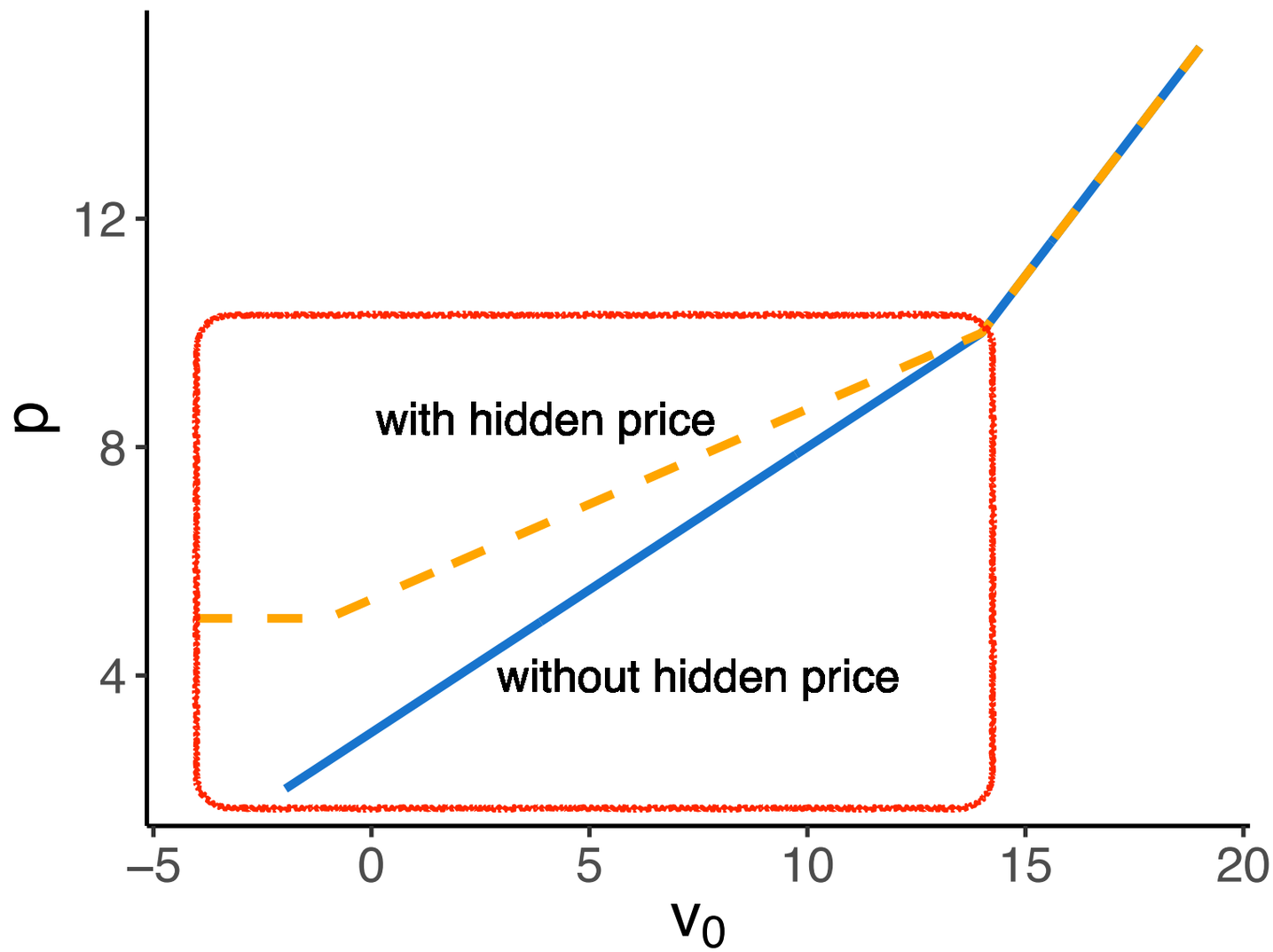


$m = 2,\ \sigma = 0.4,\ c = 0.01$

Strictly higher profit with a hidden fee in the red area



Strictly higher expected search time with a hidden fee in the red area



Heterogenous consumers

Heterogenous consumers

1. Heterogenous learning speeds

2. Heterogenous initial valuations

Heterogenous learning speeds

- Two groups of consumers with $\sigma \in \{\sigma^H, \sigma^L\}$, where $\sigma^H > \sigma^L$
- $Prob(\sigma = \sigma^H) = \rho_\sigma$
- Distribution of consumer types is common knowledge
- Realized type is each consumer’s private information
- Firm's pricing decision can lead to different strategic effects on different consumer segments
  - e.g., a higher price may generate higher profits among fast-learning consumers, but may drive slow-learning consumers out of the market

Total price with hidden fees may be lower than the price without hidden fees

- Sufficient condition:

$$\rho_\sigma < \widehat{\rho_\sigma},\ \sigma_H > \sqrt{5}\sigma_L,\ \max\{-m/2, -3\sigma_L^2/4c\} < v_0 < -\sigma_L^2/4c + m$$

- Key mechanism:

<u>Without hidden fees:</u>

Not profitable to sell to slow-learning consumers

Sellers set the price optimal for fast-learning consumers (high price)

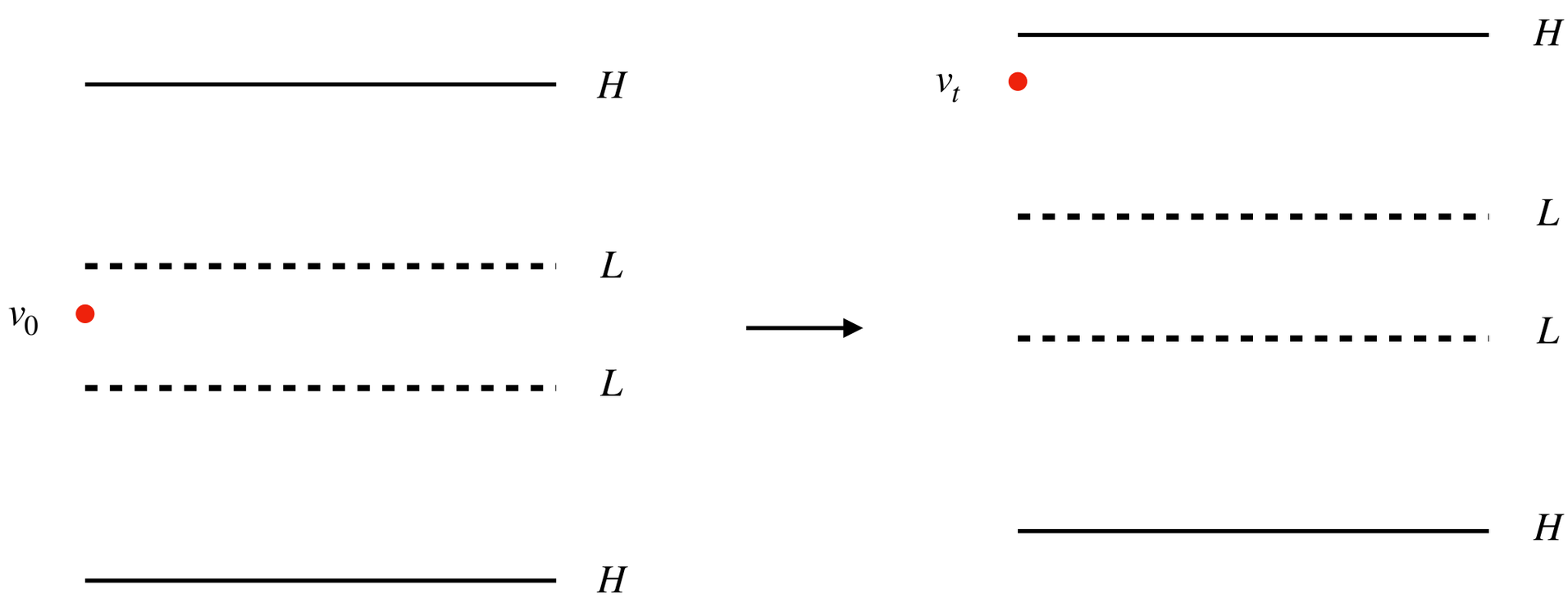
<u>With hidden fees:</u>

profitable to sell to both types of consumers

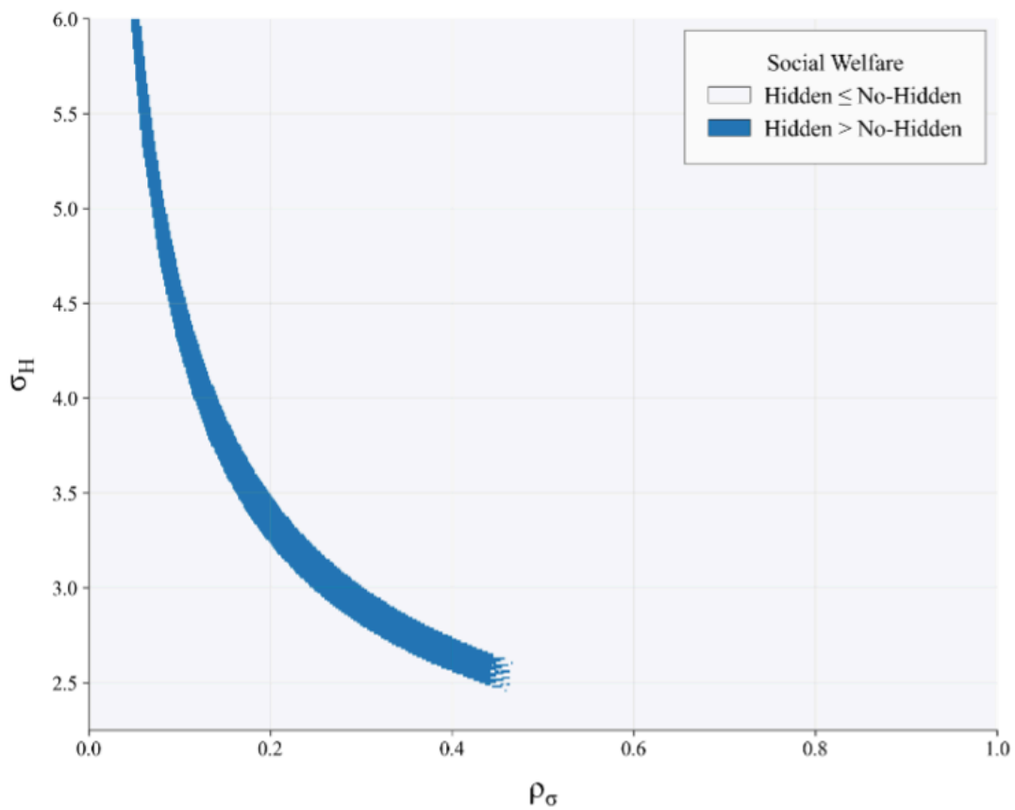
reduce the price to expand the market (low price)

Expected search time

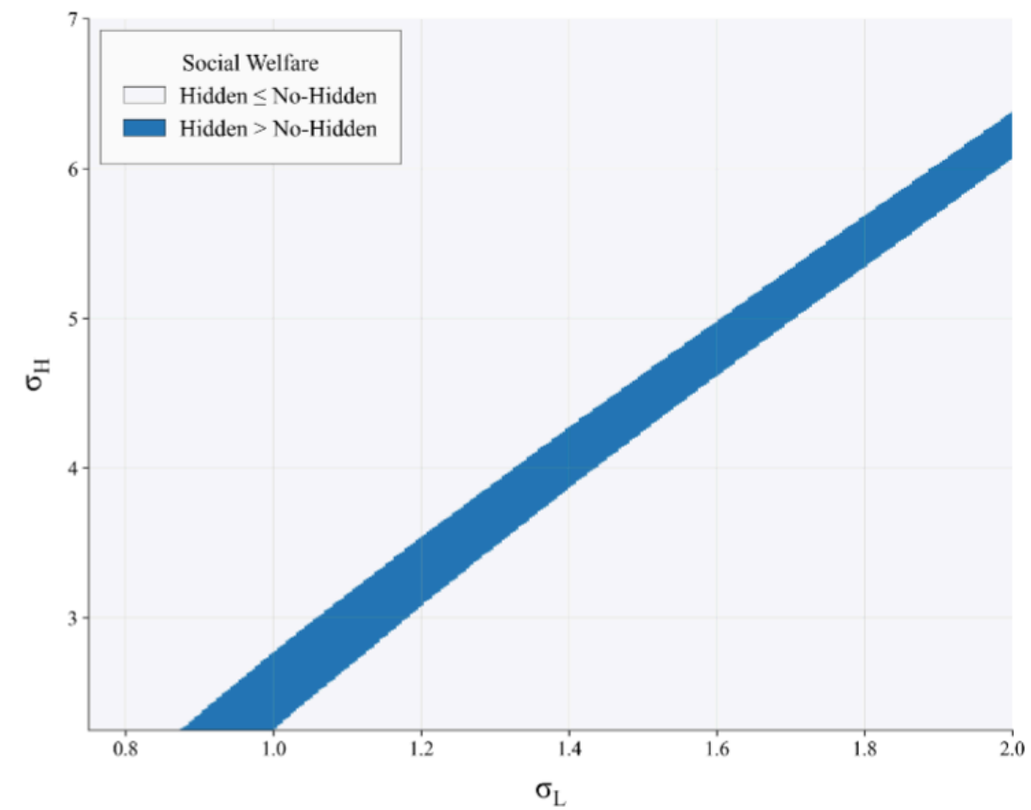
- Consumer search behavior exhibits different patterns when compared with the homogeneous consumer case
- Homogeneous case: $\mathbf{E}(\tau_1) \leq \mathbf{E}(\tau_2)$
- This case: $\mathbf{E}(\tau_1^H) > \mathbf{E}(\tau_2^H)$



The use of hidden fee can even increase social welfare



(a)



(b)

Heterogenous initial valuations

- Two groups of consumers with $v_0 \in \{v_0^H, v_0^L\}$, where $v_0^H > v_0^L$
- $Prob(v_0 = v_0^H) = \rho_v$
- Distribution of consumer types is common knowledge
- Realized type is each consumer's private information

Total price with hidden fees may be lower than the price without hidden fees

- Sufficient condition:

(1)$\rho_v < \widehat{\rho_v}$, $v_0^H - v_0^L > \sigma^2/c$, and
$\max\{-m/2, -3\sigma^2/4c + m\} < v_0^L < -\sigma^2/4c + m < v_0^H < 3\sigma^2/4c + m$

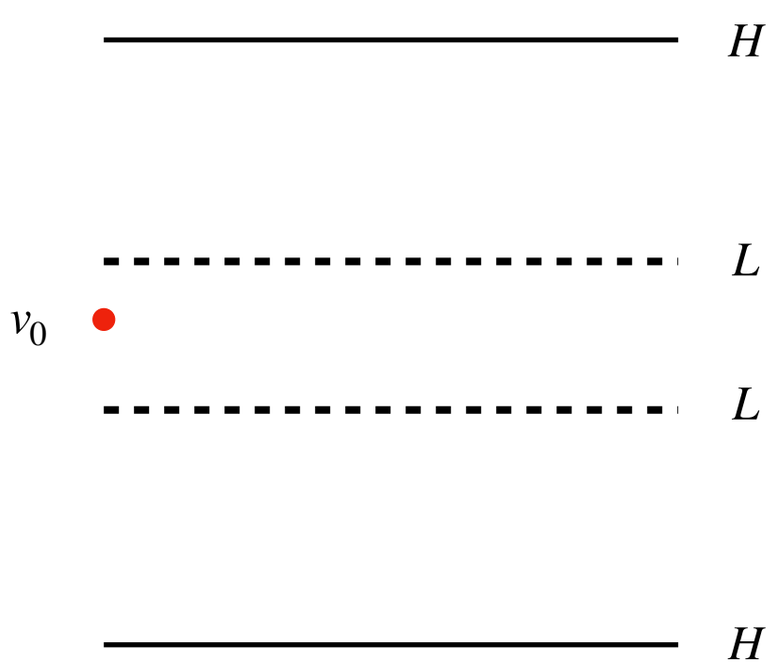
or

(2) $\rho_v < \widehat{\rho_v}'$, $v_0^H - v_0^L > \sigma^2/c$,
$-\sigma^2/4c < v_0^L < -\sigma^2/4c + m < v_0^H < 3\sigma^2/4c + m$, $v_0^L < -m/2$, and
$m < \sigma^2/2c$.

Expected search time

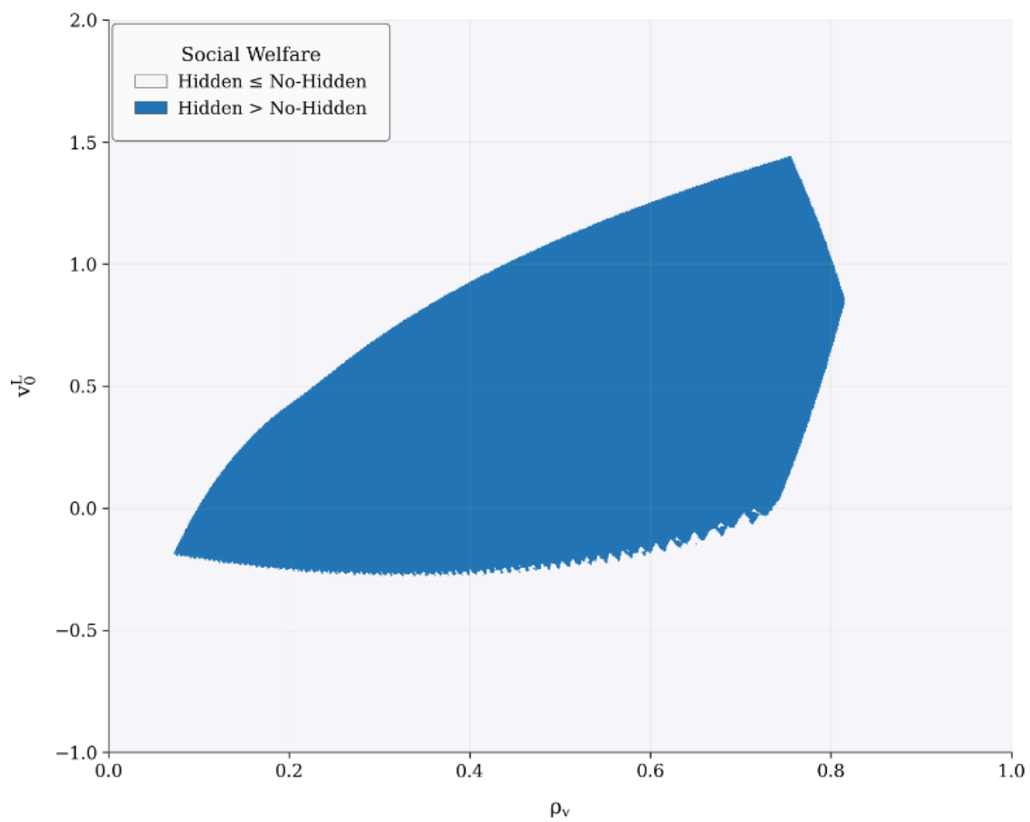
- Consumer search behavior qualitatively differ from the heterogeneous learning speeds case
- This case: $\mathbf{E}(\tau_1^H) = \mathbf{E}(\tau_2^H) = 0$

heterogeneous learning speeds

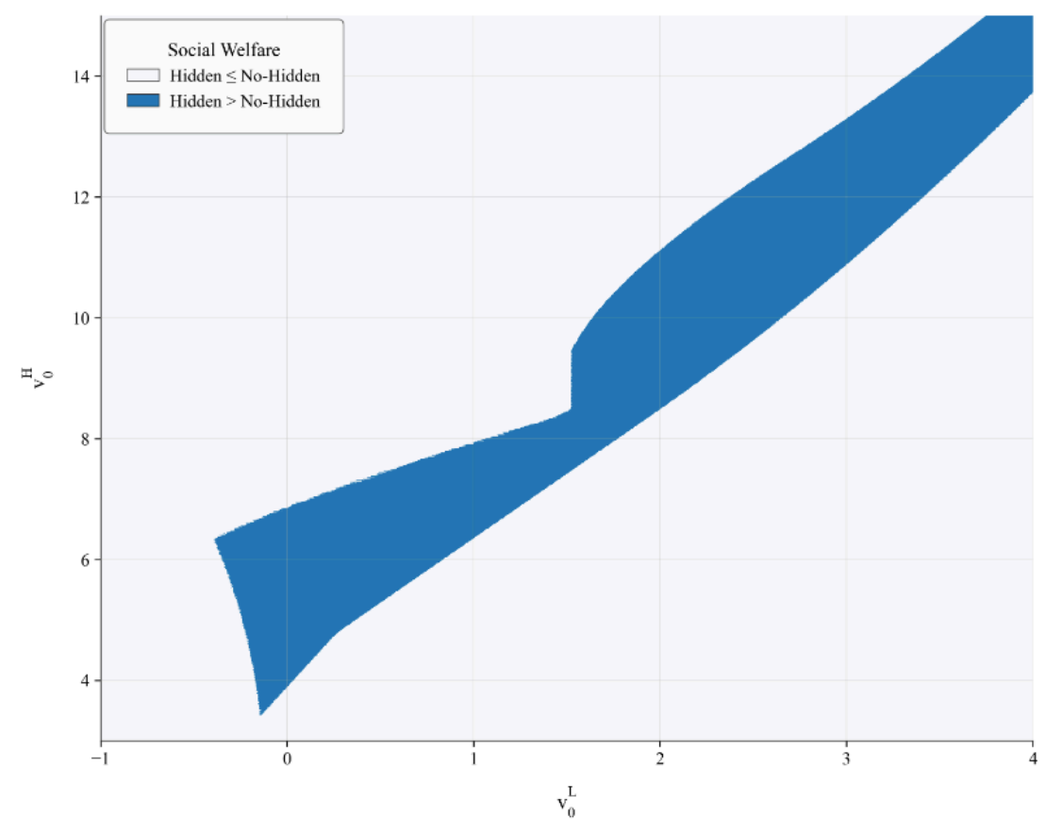


heterogeneous initial valuations

The use of hidden fee can even increase social welfare



(a)



(b)

Policy implications

Nuance Over Blanket Bans

- Hidden fees can expand market coverage and lower the price

⇒ May improve social welfare

- No one-size-fits-all approach for regulating the use of it
- Banning hidden fees in highly heterogeneous markets could backfire: can shut vulnerable consumers out of the market
- Regulators should analyze the composition of the customer base, particularly the degree of consumer heterogeneity

Extension

Some consumers are aware of the hidden fee

- ρ proportion of the consumers are unaware of the possibility of hidden fees

- Remaining $1 - \rho$ proportion of the consumers are aware of it
  $\Rightarrow$ rationally anticipate the seller's pricing strategies in the second stage

- <u>Proposition:</u> For any $\rho > 0$, there exist parameters such that the firm's expected profit is strictly higher with a hidden fee.

Thanks!